HIKITA'S CONJECTURE FOR QUIVER GAUGE THEORIES: AFTER THE SLOP

ABSTRACT. Let X be a Nakajima quiver variety, with loops and multiple arrows allowed, at a generic stability parameter, and let A be the Coulomb branch of the same gauge theory. Using the comparison map announced by Botta and Tamagni, we explain a proof that the B-algebra of A for the cocharacter given by the stability parameter is isomorphic to $H^*(X)$, equivariantly for flavor symmetries, and that the B-algebra of the quantized Coulomb branch is isomorphic to $H^*_{\mathbb{C}^\times}(X)$. The theorem and the overall strategy are due to [Ope26]. The main change is a new proof of the dimension count for quivers with a loop at every vertex, by restriction to Levi subgroups, which does not use Hausel's formula. On the quiver variety side this count becomes the decomposition theorem for the map recording the characteristic polynomials of the loops.

1. INTRODUCTION

These notes are aimed at producing a human readable and checked proof of the Hikita conjecture for quiver gauge theories. This builds on the claimed proof in the "slop dump" of Oct. 7, 2026 [Ope26], an unrefereed preprint by OpenAI dated 24 September 2026.

I should emphasize that this article is *not* a simple rewriting of the argument in that document, nor does it constitute an endorsement of the lazy approach of that dump, or the many profound violations of research ethics involved in it. It was a deeply irresponsible and selfish action carried out without regard to the consequences for individuals, or the very research apparatus that, of course, was a necessary precondition for the existence of the AI system that produced it.

However, given the existence of a claimed proof, I think it is in the interest of the mathematical community as a whole to have access to a version of the argument that is readable and checkable by humans. Given the difficulty of understanding the original argument of the OpenAI preprint, the only practical way of doing this was to use AI itself, but in this case guided by a human who understands the field. Thus, this paper was written with considerable assistance from Claude in understanding the arguments of the OpenAI preprint, finding references in the literature, and rewriting the argument in a more readable form.

[Ben: At the moment I don't want to say that humans have verified the full argument, since that isn't true. I *will*.]

Fix a finite quiver Q, loops and multiple arrows allowed, and dimension and framing vectors $\mathbf{v}, \mathbf{w} \in \mathbb{Z}_{\geq 0}^{Q_0}$. Put

$$G = \prod_{i \in Q_0} \mathrm{GL}(V_i), \qquad N = \bigoplus_{a\colon i \to j} \mathrm{Hom}(V_i, V_j) \oplus \bigoplus_i \mathrm{Hom}(W_i, V_i).$$

Let F be a torus acting on N and commuting with G, and let $\mathbb{C}^\times_\hbar$ act on T^*N with weight $\frac{1}{2}$ on both N and N^* (through its double cover), so that the symplectic form has weight 1.

For a choice of character $\nu \in \mathbb{Z}^{Q_0}$, let X_ν be the Nakajima quiver variety with this stability parameter, and let $A_{F,\hbar}$ be the quantized Coulomb branch of (G, N) with flavor symmetry F. Let $B^\nu(A_{F,\hbar})$ be the B-algebra of $A_{F,\hbar}$ for the action induced by the cocharacter ν of $\pi_1(G)^\vee$.

Theorem 1.1. *Let ν be generic. Then we have an isomorphism of graded algebras*

$$B^\nu(A_{F,\hbar}) \cong H^*_{F \times \mathbb{C}^\times_\hbar}(X_\nu).$$

The existence of a surjective map $B^\nu(A_{F,\hbar}) \to H^*_{F \times \mathbb{C}^\times_\hbar}(X_\nu)$ follows immediately from work of Botta and Tamagni [BT26]. One of the outrageous aspects of the OpenAI preprint is that

they effectively rewrote the argument of Botta and Tamagni's paper without giving them proper credit (and that OpenAI clearly did not apply sufficient due diligence to avoid this happening in many of their preprints). We only briefly summarize this construction.

Applying the graded Nakayama lemma, to complete the proof of Theorem 1.1, we only need to show an inequality

$$\dim B^\nu(A) \leq \dim H^*(X_\nu).$$

Together with the localization theorem, this lemma gives us a useful reduction: if Theorem 1.1 applies for a given choice of Q, $\mathbf{v}$, $\mathbf{w}$, F and ν, then every choice of point in the Lie algebra $\mathfrak{f}$ of F gives a vector field on X_ν whose fixed point set is a union of quiver varieties for gradings of the same quiver, and Theorem 1.1 applies to each of these fixed point components.

This brings us to one of the genuinely interesting approaches taken in the OpenAI preprint: they reduce the proof of the dimension inequality to the case of quivers with a loop at every vertex; we have consulted with many experts, and none can point to a place in the literature where this approach was used. I wish I understood the importance of this reduction better, but we find two explicit results which might help the reader see how it is helpful. In this case we have an explicit formula for a nonnegative integer $H(Q, \mathbf{v}, \mathbf{w})$ which we need to verify

(1) $\dim B^\nu(A) \leq H(Q, \mathbf{v}, \mathbf{w})$. This comes from a filtration of $B^\nu(A)$ whose graded pieces are sums, over multipartitions of $\mathbf{v}$, of quotients of the invariants of finite groups acting on the cohomology rings of hypertoric varieties (Proposition 3.6, which is [Ope26, Prop. 5.2]). Here the loops are used on the Coulomb side: they supply the roots of G as weights of N, so that the Coulomb branch is the Weyl-invariant part of an abelian Coulomb branch, with no root denominators (Proposition 3.5).

(2) $H(Q, \mathbf{v}, \mathbf{w}) \leq \dim H^*(X_\nu)$. The preprint deduces this from Hausel's formula [Hau06] for the Betti numbers of X_ν, together with an identity for the cohomology of toric varieties proved by Kirwan's method [Ope26, §6]; Remark 3.16 explains how these fit together, following notes of Nakajima [Nak26]. We give a different argument, which does not use Hausel's formula.

The filtration of $B^\nu(A)$ by subspaces F_L, indexed by the number L of parts of a multipartition, induces a filtration of $H^*(X_\nu)$ by subspaces F'_L. Thus, we only need to show that the induced map $F_L/F_{L-1} \to F'_L/F'_{L-1}$ is injective. We prove this by reduction to a Levi subgroup M of G; this gives us a new quiver variety $X(M)$, and turns the piece of F_L/F_{L-1} for a multipartition m into a level module over $H^*_M(\mathrm{pt})$. Of course, we only need to show injectivity on its socle, which is its top degree piece. Thus, we only need to show that $H^*(X(M))$ has a sufficiently large top degree piece, and ultimately we construct by hand enough distinct core components of $X(M)$ to show this.

Here

$$H(Q, \mathbf{v}, \mathbf{w}) = \sum_m \dim \left(S_m/J_m\right)^{\mathcal{W}_m},$$

the sum over multipartitions m of $\mathbf{v}$, where S_m/J_m is the cohomology ring of a hypertoric variety and $\mathcal{W}_m$ is a product of symmetric groups acting on it (§3.1). For example, for the Jordan quiver with $\mathbf{w} = 1$, so that $X_\nu = \mathrm{Hilb}^n(\mathbb{C}^2)$, every summand is 1, and $H(Q, n, 1)$ is the number of partitions of n. Both sides of (1) and (2) are independent of the generic parameter ν (Lemmas 2.1 and 4.2), so we work at determinant stability. Keeping track of degrees gives the Poincaré polynomial of X_ν as well (Theorem 3.1); Example 3.i works out a small case.

The loops can then be removed. Given Q, add a loop ℓ_k^+ at every vertex, with a flavor torus $\mathbb{C}_u^\times$ scaling it. As above, the quiver variety X_ν of Q is a component of the $\mathbb{C}_u^\times$-fixed locus of the quiver variety of the new quiver, the one where the added loops vanish. The localization theorem, and a matching statement on the Coulomb side after localizing at $u = 1$, transfer the theorem from the new quiver to Q (Proposition 4.3, which is [Ope26, Prop. 7.3]).

On the quiver variety side, the multipartitions have a direct meaning. The characteristic polynomials of the loops give a map χ from X_ν to a space of tuples of polynomials, and the

summand of m in $H(Q,\mathbf{v},\mathbf{w})$ comes from the stratum where the eigenvalues of the loops have multiplicities given by m: the decomposition theorem for χ, read at one point, gives $H^*(X_\nu)$ as a sum over multipartitions (Theorem 6.1). We do not use this in the proof.

1.1. **Set-up.** We now make the objects above precise. With $\mathbb{C}^\times_\hbar$ acting on T^*N as above, on the Coulomb side $\hbar$ is the equivariant parameter of loop rotation combined with the scaling of N with weight $\frac{1}{2}$, as in [HKW23, §§2.2, 5.7]. Other conical actions in the literature, such as that of [Kam+19, §7.2], differ from this one by a rescaling and a cocharacter of the torus scaling the framing, which shifts the flavor parameters by multiples of $\hbar$. Put $R = H^*_{G\times F}(\mathrm{pt})$ and $R_\hbar = R[\hbar]$, and let $\mu\colon T^*N \to \mathfrak{g}^*$ be the moment map.

Stability. A point of T^*N is a representation of the doubled quiver of Q on the V_i, together with maps $i_k\colon W_k \to V_k$ and $j_k\colon V_k \to W_k$ at each vertex k. Following [Cra01, §1], we regard it as a representation of dimension $(\mathbf{v},1)$ of the doubled quiver of $\hat{Q}$, which is Q with one more vertex ∞ and w_i arrows from ∞ to i. For $\nu \in \mathbb{Z}^{Q_0}$ put $\hat{\theta} = (\nu, -\nu\cdot\mathbf{v})$. A point is *$\nu$-semistable* if $\hat{\theta}\cdot\dim S \geq 0$ for every subrepresentation S, and *ν-stable* if moreover equality holds only for $S=0$ and for the whole representation [Kin94]. The same inequalities define θ-stability, and the quiver variety X_θ below, for real $\theta \in \mathbb{R}^{Q_0}$. Call ν *generic* if $\nu\cdot\mathbf{b} \neq 0$ for all $0 < \mathbf{b} \leq \mathbf{v}$. Then every ν-semistable point is stable, with trivial stabilizer in G, and the semistable locus depends only on the signs of the $\nu\cdot\mathbf{b}$. For generic ν the quiver variety $X_\nu = \mu^{-1}(0)^{\nu\text{-ss}}/G$ is smooth, and $H^*_{G\times F}(\mu^{-1}(0)^{\nu\text{-ss}}) = H^*_F(X_\nu)$. The tautological bundle $\mathcal{V}_i$ on X_ν comes from the G-module V_i. The *Kirwan map* $\kappa\colon R \to H^*_F(X_\nu)$ is restriction from $H^*_{G\times F}(\mathrm{pt})$ to $H^*_{G\times F}(\mu^{-1}(0)^{\nu\text{-ss}})$; it sends a symmetric polynomial in the Chern roots of the V_i to the same polynomial in the F-equivariant Chern roots of the $\mathcal{V}_i$. Determinant stability $\nu_{\det} = (1,\dots,1)$ is generic.

This is stability for the character $\chi_\nu = \prod_i \det_{V_i}^{\nu_i}$, with semi-invariants $f(g\cdot x) = \chi_\nu(g)^n f(x)$, as in [HKW23, §2.1]; for $\nu_{\det}$ it says that the images of the W_k generate V, by (I5). The stability condition of Nakajima, under which no nonzero invariant subspace lies in the kernel of the maps j_k, is $\nu = (-1,\dots,-1)$ in this convention [Nak98, Lemma 3.8]; compare [HKW23, Rem. 7.12].

Coulomb branches. Let $A_{F,\hbar}$ be the quantized Coulomb branch of (G,N) with flavor symmetry F [BFN18], and $A_F = A_{F,\hbar}/\hbar$, which is commutative. Both are graded by $\pi_1(G) = \mathbb{Z}^{Q_0}$ through the components of the affine Grassmannian; we call this degree the *charge*. The charge-zero part contains $R_\hbar$, respectively R, as the classes supported over the base point. For a charge c put $\deg_\nu(c) = -\sum_i \nu_i c_i$, and let $A_{[k]}$ be the sum of the charge components with $\deg_\nu = k$. The *B-algebra* is

$$B^\nu(A_{F,\hbar}) = A_{[0]} \Big/ \textstyle\sum_{k>0} A_{[-k]}A_{[k]},$$

and B^ν_F is the same construction for A_F. For $\hbar \neq 0$ the order of the factors matters: reversing it amounts to replacing ν by $-\nu$. Our ν-degree is the grading of [HKW23, §5.1.6] for the character χ_ν, and B^ν is the B-algebra of [HKW23, §5.7] and [Kam+19, §3.5]. As A_F is commutative, B^ν_F is the coordinate ring of the fixed-point scheme of the cocharacter ν of the torus $\pi_1(G)^\vee$, which acts on $\operatorname{Spec} A_F$ through the charge. The fixed-point scheme of the whole torus has coordinate ring $B_F = A_F/\langle A_F^{(c)} : c \neq 0\rangle$, where $A_F^{(c)}$ is the part of charge c; it is a quotient of B^ν_F. We write A_0, B^ν_0 and B_0 when F is trivial.

The statement. Our proof uses Theorem 2.3, the comparison map announced by Botta and Tamagni. Setting $\hbar = 0$ in Theorem 1.1 gives $B^\nu_F \cong H^*_F(X_\nu)$; more precisely, the maps $R \to B^\nu_F$ and κ are surjective with the same kernel (Theorem 4.4). This classical statement is the generic case of [Ope26, Thm. 1.1], which also allows ν on a hyperplane $\nu\cdot\mathbf{b} = 0$ when every semistable point is still stable. We do not treat that case. With F trivial it is Hikita's conjecture [Hik16]. Theorem 1.1 is its equivariant and quantized form, due to Nakajima [Kam+19, Conj. 8.9]; with the conventions of §1.1, it is [HKW23, Conj. 5.26] for quiver gauge theories.

Known cases. Hikita proved the case with F trivial for Hilbert schemes of points in the plane, type A Spaltenstein varieties and smooth hypertoric varieties [Hik16]. For ADE quivers it is [Kam+19, Thm. 8.1], whose proof uses an isomorphism of Zhu that is known in types A, D and E_6 and in some cases of types E_7 and E_8 [DK25, Rem. 2.18, Cor. 2.17]. With flavor, the classical statement is known for the Gieseker variety, the Jordan quiver with framing r [KS25, Thm. 1.14], and for type A quivers and the same ADE cases [DK25, Cors. 2.15, 2.17]. The quantized statement is known for smooth hypertoric varieties [KMP21, Thm. 1.2, Rem. 5.3]. Dinkins, Karpov and Krylov prove a refined quantum Hikita conjecture for ADE quivers framed at minuscule nodes, including all type A quivers and the Jordan quiver [DKK26, Thm. B].

1.2. **The argument.** §2 shows that $R_\hbar \to B^\nu(A_{F,\hbar})$ is onto and states the comparison map. §3 proves Theorem 3.1, the case of a loop at every vertex, at determinant stability and without flavor, by establishing (1) and (2) above. §4 adds flavor and $\hbar$ by graded Nakayama (Lemma 4.1), removes the loops (Proposition 4.3) and changes the stability parameter (Lemma 4.2); this proves Theorem 1.1 (Theorem 4.4). §6 describes §3 on the quiver variety side: the multipartition filtration becomes the support filtration of the decomposition theorem for χ. It is not used elsewhere. §5 gives a second proof of Proposition 4.3, which uses neither §3 nor added loops, but depends on results on cohomological Hall algebras whose proofs are only sketched in the literature.

Status. The argument is complete given the cited results and Theorem 2.3, which is announced in [BT26]. §7 lists what remains to be checked.

1.3. **Inputs.** We use the following results.

(I1) *Generation* [Wee19, Prop. 3.1, Rem. 3.3]. $A_{F,\hbar}$ is generated over $R_\hbar$ by the dressed minuscule monopole operators $M_{\lambda,p}$, for dominant coweights λ of G with all entries in $\{0,1\}$ or all in $\{0,-1\}$. Here the orbit Gr^λ is closed, $M_{\lambda,p}$ is the class of the part of the BFN space of triples over Gr^λ multiplied by the dressing p, and p runs over $H^*_{G\times F}(\mathrm{Gr}^\lambda) = H^*_{T\times F}(\mathrm{pt})^{\mathcal{W}_\lambda}$, with $T \subseteq G$ the diagonal torus and $\mathcal{W}_\lambda$ the stabilizer of λ in the Weyl group. The same holds for A_F over R. The proof generates the associated graded of the filtration by support in the orbit closures $\overline{\mathrm{Gr}^\Lambda}$ [BFN18, Prop. 6.8]. So the filtered piece $A_{\le\Lambda}$ is spanned by words in the generators whose coweights sum to at most Λ.

(I2) *Base change* [BFN18, Prop. 3.22]. $A_{F,\hbar}$ is free over $H_{F,\hbar} = H^*_{F\times\mathbb{C}^\times_\hbar}(\mathrm{pt})$, and $A_{F,\hbar} \otimes_{H_{F,\hbar}} \mathbb{C} = A_0$; likewise $A_{F,\hbar} \otimes_{H_{F,\hbar}} H^*_F(\mathrm{pt}) = A_F$. The flavor parameters and $\hbar$ have charge zero, and the B-algebra is a cokernel, so taking B-algebras commutes with these base changes.

(I3) *Kirwan surjectivity* [MN18, Thm. 1.2, Cor. 1.5]. κ is surjective, $H^{\mathrm{odd}}(X_\nu) = 0$, and $H^*_F(X_\nu)$ is free over $H^*_F(\mathrm{pt})$. The same holds with $F \times \mathbb{C}^\times_\hbar$ in place of F.

(I4) *Hyperkähler rotation* [BS21, Prop. 3.18]. For generic $\theta \in \mathbb{R}^{Q_0}$, X_θ is homeomorphic to the affine quotient $\mu^{-1}(-i\theta)/\!/G$. Here $\mathbb{C}^{Q_0}$ is identified with the center of $\mathfrak{g}^* \cong \mathfrak{g}$ by $\zeta \mapsto \sum_i \zeta_i \,\mathrm{id}_{V_i}$. Loops are allowed.

(I5) *Determinant stability* [HKW23, Lemma 7.1]. A point of T^*N is $\nu_{\det}$-semistable if and only if the images of the W_i generate $V = \bigoplus_i V_i$ under the arrows of the doubled quiver. Loops are allowed.

(I6) *Hypertoric varieties* [HS02]. Let a torus S act linearly on a vector space U, with weights $u_1, \dots, u_n$ (repeated according to multiplicity), none of them zero. Let $\mu\colon T^*U \to \mathrm{Lie}(S)^*$ be the moment map and θ a generic character of S. The *hypertoric variety* is $\mu^{-1}(0)/\!/_\theta S$. A *cocircuit* is a minimal set of weights whose complement does not span the span of all the u_i. Suppose the u_i span $X^*(S) \otimes \mathbb{Q}$. Then:

(a) if the u_i are unimodular, the hypertoric variety is smooth, of dimension $2(n - \dim S)$ when it is nonempty [HS02];

(b) its cohomology ring is $H^*_S(\mathrm{pt})$ modulo the ideal generated by the products $\prod_{u\in C} u$ over the cocircuits C [HS02, Thm. 1.1];
(c) this ring vanishes above degree $n - \dim S$, and it is level in the sense of (I8)(c) [HS02, Thm. 7.1, Prop. 6.7], [Sta83, Thm. III.3.4];
(d) the top Betti number is the top h-number of the matroid of the arrangement [HS02, Cor. 8.4], which is dual to the matroid of the u_i; so it is the cographic matroid when the u_i are the vectors of a graph. For a matroid $\mathcal{M}$ of rank r the top h-number is $T_{\mathcal{M}}(0,1)$, and $T_{\mathcal{M}^*}(x,y) = T_{\mathcal{M}}(y,x)$, where T is the Tutte polynomial.

(I7) *Cuts of a graph.* Let Γ be a connected graph, possibly with parallel edges, with one vertex ∞ marked. For a set I of vertices, the *cut* $\delta(I)$ is the set of edges with exactly one end in I. A *bond* is a minimal nonempty cut.
(a) Give the edge between D and E the vector $x_D - x_E$, and the edge between D and ∞ the vector x_D. This is the usual representation of the graphic matroid, with $x_\infty = 0$. Its rank is the number of vertices other than ∞, its cocircuits are the bonds, and every bond is $\delta(I)$ for some I not containing ∞.
(b) Every cut is a disjoint union of bonds.
(c) The vectors of (a) are unimodular [HS02, Lemma 8.1].
(d) The number of acyclic orientations of Γ with ∞ the only source is $T_\Gamma(1,0)$ [GZ83], where T_Γ is the Tutte polynomial.

(I8) *Finite groups and graded algebras.*
(a) Let a finite group H act linearly on a complex vector space U. The average $\mathrm{Av}_H = |H|^{-1}\sum_{h\in H} h$ is an H-equivariant projection of U onto U^H. So for an H-stable subspace $U' \subseteq U$ we have $\mathrm{Av}_H(U') = U'^H$ and $(U/U')^H = U^H/U'^H$.
(b) The symmetric polynomials in n variables are generated by the power sums of degrees $1,\dots,n$ (Newton's identities).
(c) Let $C = \bigoplus_{i\ge 0} C_i$ be a finite-dimensional graded commutative algebra with $C_0 = \mathbb{C}$, and $C_+ = \bigoplus_{i>0} C_i$. The *socle* of C is the annihilator of C_+, and C is *level* if its socle is its top nonzero degree. Every nonzero graded ideal K of C meets the socle.

(I9) *Sylvester's equation.* Let $A \in \mathrm{End}(U)$ and $B \in \mathrm{End}(U')$ have disjoint spectra. Then $X \mapsto AX - XB$ is an invertible map of $\mathrm{Hom}(U', U)$, and the solution of $AX - XB = C$ is a linear combination of maps $A^r C B^s$. Similarly, if $U = \bigoplus_D U_D$ and $A = \bigoplus_D A_D$ with pairwise disjoint spectra, then the projection of U onto each U_D is a polynomial in A.

(I10) *Cores* [HV13]. Let X be a quiver variety at a generic parameter ν, as in §1.1. The map $\pi\colon X \to \mu^{-1}(0)/\!/G$ to the affine quotient is projective. The *core* of X is the projective variety $\mathfrak{L} = \pi^{-1}(\pi(0))$. Let $\mathbb{C}^\times$ scale T^*N with weight one. This preserves $\mu^{-1}(0)$ and ν-stability, so it acts on X, and for every $x \in X$ the limit $\lim_{t\to 0} t\cdot x$ exists in X and lies in $\mathfrak{L}$.
(a) X retracts onto $\mathfrak{L}$, so restriction $H^*(X) \to H^*(\mathfrak{L})$ is an isomorphism [HV13].
(b) Every G-invariant function on $\mu^{-1}(0)$ that vanishes at 0 vanishes on (the preimage of) $\mathfrak{L}$. For example, for every loop ℓ of the quiver, the traces of the powers of ℓ vanish on $\mathfrak{L}$, so ℓ is nilpotent there.
(c) If $p \in \mu^{-1}(0)$ is stable and $\lambda(t)\cdot p \to 0$ as $t \to 0$ for a cocharacter λ of G, then $[p] \in \mathfrak{L}$.
(d) If $\dim \mathfrak{L} = e$, then $\dim H^{2e}(\mathfrak{L})$ is the number of irreducible components of $\mathfrak{L}$ of dimension e [CG97, §2.6].

(I11) *Weights and the decomposition theorem*, used only in §6. Complexes of sheaves on complex varieties are taken in the derived category of mixed Hodge modules [Sai90], or, after spreading out, of mixed ℓ-adic complexes [BBD82]. Perverse sheaves are normalized so that a local system $\mathcal{L}$ on a smooth variety Y gives the perverse sheaf $\mathcal{L}[\dim Y]$, and $\mathrm{IC}(\overline{Y}, \mathcal{L})$ is its intermediate extension to $\overline{Y}$.

(a) A complex K has weights $\leq w$ if and only if each stalk $H^i(K_x)$ has weights $\leq w+i$ (the definition [BBD82, §5.1.8]). For f a morphism, Rf_* preserves weights $\geq w$ [BBD82, §5.1.14]. On a smooth variety, $\mathbb{Q}$ is pure of weight 0. If X is smooth and $H^*(X) \to H^*(\mathfrak{L})$ is an isomorphism for some proper $\mathfrak{L} \subseteq X$, then $H^i(X)$ is pure of weight i: smoothness gives weights $\geq i$, and properness of $\mathfrak{L}$ gives weights $\leq i$.

(b) A pure complex is, after extending scalars to an algebraic closure in the ℓ-adic case, a direct sum of shifted simple perverse sheaves $\mathrm{IC}(\overline{Y}, \mathcal{L})$, with Y smooth irreducible and $\mathcal{L}$ an irreducible local system on Y [BBD82, §§5.3.8, 5.4.5], [Sai90].

(c) Let y be a point of $\overline{Y}$ with $\dim Y > 0$, such that $\mathrm{IC}(\overline{Y}, \mathcal{L})$ is constructible for a stratification having $\{y\}$ as a stratum, and let i_y be its inclusion. Then $H^j(i_y^* \mathrm{IC}(\overline{Y}, \mathcal{L})) = 0$ for $j \geq 0$ and $H^j(i_y^! \mathrm{IC}(\overline{Y}, \mathcal{L})) = 0$ for $j \leq 0$. So the natural map $i_y^! \to i_y^*$ is zero on it.

(d) Let $\mathbb{C}^\times$ act on an affine space V with positive weights, and let K be constructible for a stratification by $\mathbb{C}^\times$-stable strata. Then restriction $R\Gamma(V, K) \to K_0$ is an isomorphism [KS94, Prop. 3.7.5].

(e) Let a finite group W act linearly on $\mathbb{A}^l$, with quotient $q\colon \mathbb{A}^l \to \mathbb{A}^l/W$, and let $\mathcal{L}(V)$ be the local system given by a W-module V on the image of the free locus. Then $\mathrm{IC}(\mathbb{A}^l/W, \mathcal{L}(V)) = (q_* \mathbb{Q}[l] \otimes V)^W$. Its stalk at $q(0)$ is V^W in degree $-l$, and the restriction from this stalk to the stalk at a free point is the inclusion $V^W \subseteq V$.

2. The comparison map

We regard $R_\hbar$ as the subalgebra of $A_{F,\hbar}$ of classes supported over the base point. For generic ν the map $R_\hbar \to B^\nu(A_{F,\hbar})$ is onto. At $\hbar = 0$ this is elementary, and we can say more:

Lemma 2.1. *Let ν be generic, let $A_{[k]} \subseteq A_F$ be the part of ν-degree k, and let $I_B \subseteq A_F$ be the ideal generated by the elements of nonzero charge. Put*

$$J_\nu = \sum_{k>0} A_{[-k]} A_{[k]}.$$

Then $A_{[0]} = R + J_\nu$ and $I_B \cap A_{[0]} = J_\nu$. So $R \to B_F^\nu$ is onto, and $B_F^\nu = B_F$.

Proof. The charge of a coweight λ of G is the vector of the sums of its entries at each vertex. So a generator $M_{\lambda,p}$ of (I1) with $\lambda \neq 0$ has a charge c with $0 < c \leq \mathbf{v}$ or $0 < -c \leq \mathbf{v}$, and $\deg_\nu(c) \neq 0$ as ν is generic. The generators with $\lambda = 0$ are the elements of R. Now $A_{[0]}$ is spanned by the words in the generators whose ν-degrees add up to zero. If such a word has a letter of nonzero charge, then some letter has degree $-k < 0$, and moving it to the front shows that the word lies in $A_{[-k]} A_{[k]}$, as A_F is commutative. Otherwise the word lies in R. So $A_{[0]} = R + J_\nu$. The same argument shows that every charge component $A^{(c)}$ with $c \neq 0$ and $\deg_\nu(c) = 0$ lies in J_ν, since each word of charge c has a letter of nonzero charge.

Decomposing I_B by charge, this gives $I_B \cap A_{[0]} = J_\nu$. As $A_F = A^{(0)} + I_B$ and $A^{(0)} \subseteq A_{[0]}$, the map $B_F^\nu = A_{[0]}/J_\nu \to A_F/I_B = B_F$ is an isomorphism. □

For $\hbar \neq 0$ the algebra $A_{F,\hbar}$ is not commutative, and we reorder:

Lemma 2.2. *Let ν be generic, and let $A_{[k]} \subseteq A_{F,\hbar}$ be the part of ν-degree k. Then*

$$A_{[0]} = R_\hbar + J, \qquad J = \sum_{k>0} A_{[-k]} A_{[k]},$$

so $R_\hbar \to B^\nu(A_{F,\hbar})$ is onto.

Proof. Fix a dominant coweight Λ. The filtered piece $A_{\leq \Lambda}$ of (I1) is a finitely generated graded $R_\hbar$-module, so the degrees of $A_{\leq \Lambda} \cap A_{[0]}$ are bounded below, and we show that every homogeneous element of it lies in $R_\hbar + J$ by induction on degree. By (I1), such an element is a combination of words $y = m_1 \cdots m_r$ in the generators, of the same degree and of ν-degree zero,

whose coweights sum to at most Λ. If no letter has nonzero charge, then $y \in R_\hbar$. Otherwise, as in the proof of Lemma 2.1, some letter m_j has ν-degree $-k < 0$, and

$$y = m_j \cdot m_1 \cdots \widehat{m_j} \cdots m_r + \sum_{i<j} m_1 \cdots m_{i-1} [m_i, m_j] m_{i+1} \cdots \widehat{m_j} \cdots m_r.$$

The first term lies in $A_{[-k]}A_{[k]}$. Since A_F is commutative, $[m_i, m_j] = \hbar c$ for some c, and c lies in the same filtered piece as $[m_i, m_j]$, because the subquotients of the filtration are free over $\mathbb{C}[\hbar]$ [BFN18, §6(i)]. So each other term is $\hbar$ times an element of $A_{\le\Lambda} \cap A_{[0]}$ of lower degree, which lies in $R_\hbar + J$ by induction. □

The comparison map is the following.

Theorem 2.3 (Botta–Tamagni, announced)**.** *For every generic ν, the surjective maps $R_\hbar \to B^\nu(A_{F,\hbar})$ and $R_\hbar \to H^*_{F\times\mathbb{C}^\times_\hbar}(X_\nu)$ induce a surjective homomorphism of graded $R_\hbar$-algebras*

$$\Phi\colon B^\nu(A_{F,\hbar}) \to H^*_{F\times\mathbb{C}^\times_\hbar}(X_\nu).$$

This is announced in [BT26, Thm. E, §0.1.6.4], as a consequence of the Coulomb branch action on the critical cohomology of based quasimaps constructed there. We use it without giving a proof.

Setting $\hbar = 0$, and also the flavor parameters to zero, gives surjections

$$\Phi\colon B^\nu_F \to H^*_F(X_\nu), \qquad \Phi\colon B^\nu_0 \to H^*(X_\nu),$$

with $\Phi[f] = \kappa(f)$ for $f \in R$. Indeed, the B-algebras specialize by (I2), and so does the equivariant cohomology, as it is free over $H^*_{F\times\mathbb{C}^\times_\hbar}(\mathrm{pt})$ by (I3).

3. Quivers with a loop at every vertex

In this section F is trivial, $\hbar = 0$, every vertex k with $v_k > 0$ carries a loop, and $\nu = \nu_{\det}$. Write $A = A_0$, $B = B_0$ and $X = X_{\nu_{\det}}$, and let $I_B \subseteq A$ be the ideal generated by the elements of nonzero charge, so that $B = A/I_B$.

As $\nu_{\det}$ is generic, Lemma 2.1 gives $B_0^{\nu_{\det}} = B$, and $R \to B$ is onto. By §2 we have a surjection $\Phi\colon B \to H^*(X)$ with $\Phi[f] = \kappa(f)$ for $f \in R$. We grade by half the cohomological degree.

Theorem 3.1. *For every quiver with a loop at every vertex of positive dimension, $\Phi\colon B \to H^*(X)$ is an isomorphism, and the Poincaré polynomial $P_X(q) = \sum_i \dim H^{2i}(X)\, q^i$ of X is*

$$P_X(q) = \sum_m q^{\delta_m} h_m(q),$$

the sum over multipartitions m, with δ_m and h_m as in §3.1.

[Ope26, Prop. 6.4] derives this formula for P_X from Hausel's formula [Hau06]. Here it comes out of the proof; Remark 3.16 explains how the two are related, and Example 3.i works out a small case. For the Jordan quiver with $\mathbf{w} = 1$, so that $X = \mathrm{Hilb}^n(\mathbb{C}^2)$, Q' has no arrows, each S_m/J_m is $\mathbb{C}$, and $\delta_m = n - l(m)$; the formula is $P_X(q) = \sum_{m\vdash n} q^{n-l(m)}$. For $\mathbf{w} = r$, X is the Gieseker variety $M(r,n)$; now Γ_m has r edges from each part to ∞ and none between parts, so $S_m/J_m = \bigotimes_D \mathbb{C}[x_D]/(x_D^r)$ is the cohomology of $(\mathbb{P}^{r-1})^{l(m)}$, and $\delta_m = r(n-l(m))$. If m has a_s parts of size s, then $\mathcal{W}_m = \prod_s \mathfrak{S}_{a_s}$, the invariants are the cohomology of $\prod_s \mathrm{Sym}^{a_s}\mathbb{P}^{r-1}$, and

$$P_{M(r,n)}(q) = \sum_{m\vdash n} q^{r(n-l(m))} \prod_s \binom{a_s + r - 1}{a_s}_q,$$

with q-binomial coefficients. For $r = 2$ and $n = 2$ this is $(1+q+q^2)+q^2(1+q) = 1+q+2q^2+q^3$.

Plan. We follow the outline in the introduction.

(a) *The filtration* (§3.3). Proposition 3.6 filters B by subspaces F_L, indexed by the number L of parts of a multipartition, with surjections $\bigoplus_{l(m)=L}(S_m/J_m)^{\mathcal{W}_m}\langle\delta_m\rangle \twoheadrightarrow F_L/F_{L-1}$, where S_m/J_m is the cohomology of a hypertoric variety (Proposition 3.3). So $\dim B \le H(Q,\mathbf{v},\mathbf{w})$.

(b) *The induced filtration* (§3.4). The images $F'_L = \Phi(F_L)$ filter $H^*(X)$, and it suffices to show that the map from $\bigoplus_{l(m)=L}(S_m/J_m)^{\mathcal{W}_m}\langle\delta_m\rangle$ to F'_L/F'_{L-1} is injective (Lemma 3.7).

(c) *Levi subgroups* (§3.5). Restriction to the Levi subgroup M_m, on the Coulomb side, and pullback along an étale map $X(M_m)^\circ \to X$, on the quiver variety side, turn the summand of m into the piece S_m/J_m of the multipartition with one part per block for the quiver Q_{M_m}, with no invariants taken (Proposition 3.10). So it suffices to treat that piece for every quiver with a loop at every vertex.

(d) *The socle and the top degree* (§3.6). That piece is a level module over R, so it suffices to check injectivity on its socle, its top degree (Lemma 3.11). There we need enough classes in the top degree of $H^*(X)$, and we construct by hand enough components of the core of X (Lemma 3.13, Corollary 3.14).

§3.7 puts these together.

3.1. **Notation.** Let $Q_0^+ = \{k : v_k > 0\}$ and $k_0 = |Q_0^+|$. Choose a loop ℓ_k at each $k \in Q_0^+$, and let Q' be Q with these loops removed. Put $N' = N_{Q'} \oplus \operatorname{Hom}(W,V)$, so $N = \mathfrak{g} \oplus N'$. Let c_{kl} be the number of arrows of Q' between k and l, in either direction, so that a loop of Q' at k counts twice in c_{kk}. Put $z = \frac{1}{2}\sum_k c_{kk}v_k$, the number of zero weights of the diagonal torus on N'. A point of T^*N has two components in $\operatorname{End}(V_k)$ for the chosen loop ℓ_k: the map along the loop, which we also call ℓ_k, and the map along the reversed loop, coming from the doubling of the quiver, which we call ℓ_k^*. So $\ell_k \in \mathfrak{gl}(V_k)$ is the coordinate on N, and ℓ_k^* is the dual coordinate on N^*.

Multipartitions and Levi subgroups. A *multipartition* m is a tuple of partitions

$$m^{(k)} = (m_1^{(k)} \ge \cdots \ge m_{r_k}^{(k)})$$

of v_k, $k \in Q_0^+$, where r_k is the length of $m^{(k)}$. Let $P_m = \{(k,i) : k \in Q_0^+,\ 1 \le i \le r_k\}$, the set indexing the parts of m. For $D = (k,i) \in P_m$ put $k(D) = k$ and $s_D = m_i^{(k)}$, and put $l(m) = |P_m|$; we call the elements of P_m the *parts* of m. A decomposition

$$V_k = \bigoplus_{D\in P_m,\ k(D)=k} V^{(D)}, \qquad \dim V^{(D)} = s_D,$$

gives a Levi subgroup $M_m = \prod_{D\in P_m}\operatorname{GL}(V^{(D)})$. Every Levi subgroup of G is conjugate to M_m for exactly one m.

Toric data. Let $Z_m \cong (\mathbb{C}^\times)^{l(m)}$ be the center of M_m. The group $\mathcal{W}_m = N_G(M_m)/M_m$ permutes the parts of equal size at each vertex. Let ξ_m be a regular nilpotent element of $\operatorname{Lie}(M_m)$, so that as an element of $\mathfrak{g}$ it has Jordan type m. The multipartition m_G, with one part at each vertex, has $M_{m_G} = G$.

Let $N'^{\xi_m} \subseteq N'$ be the subspace killed by ξ_m. The torus Z_m commutes with ξ_m, so it acts on N'^{ξ_m}. Write $x_D \in H^2_{Z_m}(\mathrm{pt})$ for the character by which Z_m acts on $V^{(D)}$, so that $S_m = H^*_{Z_m}(\mathrm{pt}) = \mathbb{C}[x_D : D \in P_m]$. The space N'^{ξ_m} is the sum of the following pieces, on each of which Z_m acts by a single weight.

- For each arrow from k to l of Q', and parts D at k and E at l, the space $\operatorname{Hom}(V^{(D)}, V^{(E)})^{\xi_m}$ of maps commuting with ξ_m, of dimension $\min(s_D, s_E)$, with weight $x_E - x_D$. The weight is zero exactly when $D = E$, for a loop of Q'.
- For each part D, the space $\operatorname{Hom}(W_{k(D)}, V^{(D)} \cap \ker\xi_m)$, of dimension $w_{k(D)}$, with weight x_D.

Definition 3.2. *Let Γ_m be the graph whose vertices are the parts of m and one more vertex ∞. It has $c_{k(D)k(E)}\min(s_D,s_E)$ edges between parts $D \ne E$, and $w_{k(D)}$ edges between D and*

∞. Label the edges between D and E by $x_D - x_E$, and those between D and ∞ by x_D. Up to sign, these labels are the nonzero Z_m-weights of N'^{ξ_m}, one for each weight vector. For a nonempty set I of parts, let $q_I \in S_m$ be the product of the labels of the edges with exactly one end in I, and let $J_m \subseteq S_m$ be the ideal generated by the q_I. Put

$$h_m = \mathrm{Hilb}\big((S_m/J_m)^{\mathcal{W}_m}\big), \qquad \delta_m = \dim N' - \dim N'^{\xi_m},$$

where Hilb *is the Hilbert series in q, with each x_D in degree 1.*

For a graded space U, $U\langle\delta\rangle$ is U with all degrees raised by δ.

Proposition 3.3. *We have a ring isomorphism $S_m/J_m \cong H^*(\mathfrak{M}_m)$, where $\mathfrak{M}_m$ is the hypertoric variety of Z_m acting on N'^{ξ_m} with its zero weights removed, at a generic parameter. It vanishes above degree $n_m - l(m)$, where n_m is the number of edges of Γ_m, and $\delta_m + n_m - l(m) = \dim N' - z - l(m)$. If it is nonzero, it is level: its socle is its top degree $n_m - l(m)$. That degree has dimension the number of acyclic orientations of Γ_m in which ∞ is the only source.*

Proof. By (I7)(a), with $x_\infty = 0$, the weights of Z_m on N'^{ξ_m} are the vectors of the graphic matroid of Γ_m. Suppose first that Γ_m is connected. Then they span $H^2_{Z_m}(\mathrm{pt})$, and they are unimodular by (I7)(c). By (I6)(b), $H^*(\mathfrak{M}_m)$ is S_m modulo the products of the weights over the cocircuits. By (I7)(a) the cocircuits are the bonds, and each bond is $\delta(I)$ for a nonempty set I of parts. So the ideal is contained in J_m. Conversely, each q_I is the product over the cut $\delta(I)$, which by (I7)(b) is a disjoint union of bonds, so q_I is a multiple of a bond product. So the ideal is J_m. (This is also [HS02, Thm. 8.3], which presents the cohomology of a toric quiver variety by the products over all cuts.) If Γ_m is disconnected, then a component I not containing ∞ has $q_I = 1$, so $S_m/J_m = 0$. The weights do not span, and a generic θ lies outside their span, so $\mathfrak{M}_m$ is empty and the isomorphism still holds.

By (I6)(a), $\mathfrak{M}_m$ is smooth, since the weights are unimodular, and it has dimension $2(n_m - l(m))$ when it is nonempty. By (I6)(c), with $\dim Z_m = l(m)$ and n_m nonzero weights, the ring vanishes above degree $n_m - l(m)$, and it is level.

For the formula, the edges of Γ_m correspond to the weight vectors of N'^{ξ_m} with nonzero weight, and there are z zero weights, so $n_m = \dim N'^{\xi_m} - z$ and $\delta_m + n_m = \dim N' - z$. By (I6)(d) the top Betti number is the top h-number of the cographic matroid of Γ_m, which is $T_{\Gamma_m}(1,0)$. By (I7)(d) this counts the acyclic orientations of Γ_m in which ∞ is the only source. □

Remark 3.4 (the cuts as Euler classes). The ideal J_m has a direct geometric meaning, which we learned from [Nak26, §§1–2], notes on [Ope26, §6] written by Claude at the request of H. Nakajima. Let E_m be the space of representations, with dimension one at every vertex, of the quiver with vertices $P_m \sqcup \{\infty\}$, one arrow in each direction for each edge of Γ_m between two parts, and one arrow $\infty \to D$ for each edge between D and ∞. The torus Z_m acts on E_m, trivially at ∞, with weight $x_E - x_D$ on an arrow $D \to E$ and x_D on an arrow $\infty \to D$. Let $U_m \subseteq E_m$ be the open set where every part can be reached from ∞ along nonzero arrows. As in (I5), it is the semistable locus for the determinant character, and Z_m acts freely on it, so U_m/Z_m is a smooth toric variety.

(1) Restriction $S_m = H^*_{Z_m}(E_m) \to H^*_{Z_m}(U_m) = H^*(U_m/Z_m)$ is onto, with kernel J_m. This is Kirwan's method [Kir84] for the torus Z_m and the determinant character, applied to the stratification of E_m by the set I of parts that cannot be reached; the class of the stratum of I is $\pm q_I$.

(2) Adding arrows $D \to \infty$ dual to the framing arrows turns E_m into the cotangent space of the representation space of an orientation of Γ_m. The new arrows do not help to reach parts from ∞, so the quotient of its semistable locus by Z_m, which is the Lawrence toric variety of [HS02], is a vector bundle over U_m/Z_m. For the determinant character it contains $\mathfrak{M}_m$ as the zero set of the moment map, and $\mathfrak{M}_m$ is a deformation retract

of it [HS02, Remark after the proof of Thm. 1.1]. So (1) gives the ring isomorphism of Proposition 3.3 again, with all cuts at once. In this description, $\mathfrak{M}_m$ is the quiver variety of an orientation of Γ_m, with dimension one at each part and the edges at ∞ as framing, at determinant stability.

(3) Compare Proposition 3.6(3) below. There, the cut relation for I comes from splitting off the coweight supported on the first elements of the parts in I, whose nonzero charge kills it in B, and the coefficient of r_{λ_m} in the resulting product is $\pm q_I$. Here q_I is the class of the stratum where exactly the parts in I are unreachable.

(4) The group $\mathcal{W}_m$ permutes the strata, and taking invariants in the split Thom–Gysin sequences gives

$$\frac{1}{(q)_m} = \sum_I q^{\deg q_I} \frac{h_{m'}(q)}{(q)_I},$$

the sum over the $\mathcal{W}_m$-orbits of subsets $I \subseteq P_m$, where m' is the multipartition formed by the parts outside I. Here $1/(q)_m = \prod_{k,s} \prod_{j=1}^{a_{k,s}} (1-q^j)^{-1}$ is the Hilbert series of $S_m^{\mathcal{W}_m}$, where $a_{k,s}$ is the number of parts of m of size s at k, and $(q)_I$ is defined in the same way from the parts in I. The term of $I = \emptyset$ is h_m, so this determines h_m recursively. It is [Ope26, Prop. 6.2]; see [Nak26, Prop. 2.1].

Nothing in this remark is used in the proof of Theorem 3.1.

Example 3.i. Let Q be the quiver $1 \to 2$ with a loop added at each vertex, so that Q' is $1 \to 2$, $c_{12} = 1$ and $c_{11} = c_{22} = 0$, and take $\mathbf{v} = (2,1)$ and $\mathbf{w} = (1,0)$. Then $\dim N' = v_1v_2 + w_1v_1 = 4$, $z = 0$ and $k_0 = 2$. There are two multipartitions.

- $m_G = ((2),(1))$, with a part D of size 2 at 1 and a part E of size 1 at 2. The graph Γ_{m_G} is the path $\infty - D - E$, as $w_1 = 1$, $w_2 = 0$ and $\min(2,1) = 1$. Up to sign, $q_{\{D\}} = x_D(x_D - x_E)$, $q_{\{E\}} = x_E - x_D$ and $q_{\{D,E\}} = x_D$, so $J_{m_G} = (x_D, x_E)$ and $h_{m_G} = 1$. Also $\dim N'^{\xi_{m_G}} = \min(2,1) + w_1 = 2$, so $\delta_{m_G} = 2$.
- $m = ((1,1),(1))$, with parts D_1, D_2 at 1 and E at 2, all of size one. Then $\xi_m = 0$, so $\delta_m = 0$, and Γ_m is the 4-cycle $\infty - D_1 - E - D_2 - \infty$. With $a = x_{D_1}$, $b = x_{D_2}$ and $c = x_E$, the q_I are, up to sign, $a(a-c)$, $b(b-c)$, $(c-a)(c-b)$, $a(c-b)$, $b(c-a)$, ab and $ab(a-c)(b-c)$. The first six generate $(a,b,c)^2$, so $J_m = (a,b,c)^2$, and S_m/J_m has basis $1, a, b, c$. Its top degree $n_m - l(m) = 1$ has dimension 3, the number of acyclic orientations of the 4-cycle with ∞ the only source, and $\mathfrak{M}_m$ is the minimal resolution of $\mathbb{C}^2/(\mathbb{Z}/4)$. The group $\mathcal{W}_m = \mathfrak{S}_2$ swaps a and b, so $h_m = 1 + 2q$.

By Theorem 3.1, the 8-dimensional quiver variety X has $P_X(q) = (1+2q) + q^2$, with top degree $d_{\max} = \dim N' - z - k_0 = 2$. This example is taken from [Nak26, Exs. 1.13, 3.8]. ♣

3.2. **The Coulomb branch in abelianized form.** Let $T \subseteq G$ be the diagonal torus. Index the standard bases of the V_k by $\Omega = \{(k,a) : k \in Q_0,\ 1 \le a \le v_k\}$, and write Ω_k for the elements with first entry k, so $|\Omega_k| = v_k$. So $T = (\mathbb{C}^\times)^\Omega$, coweights are elements of $\mathbb{Z}^\Omega$, and the charge of λ is $(\sum_{d\in\Omega_k} \lambda_d)_k$. Also $S = H_T^*(\mathrm{pt}) = \mathbb{C}[x_d : d \in \Omega]$, and $\mathcal{W} = \prod_k \mathfrak{S}_{v_k}$. Let $\mathcal{B}'$ be the multiset of T-weights of N'. In the commutative ring $\mathrm{Frac}(S)[\mathbb{Z}^\Omega]$, with basis e^λ over $\mathrm{Frac}(S)$, put

$$r_\lambda = e^\lambda \prod_{\beta\in\mathcal{B}'} \beta^{(-\langle\beta,\lambda\rangle)_+},$$

where $y_+ = \max(y,0)$ and $\langle\ ,\ \rangle$ pairs weights with coweights. The group $\mathcal{W}$ acts on this ring by permuting the x_d and the coordinates of λ together.

Proposition 3.5 (Teleman). *$A = \left(\bigoplus_{\lambda\in\mathbb{Z}^\Omega} S\, r_\lambda\right)^{\mathcal{W}}$, compatibly with the abelianization embedding of [BFN18, Lemma 5.17].*

This is [Tel21, App. A, Thm. 6]: if the weights of N contain the roots, the Coulomb branch is the $\mathcal{W}$-invariant part of the abelian Coulomb branch of T acting on N with the roots removed. The chosen loops supply the roots: the weights of ℓ_k are the $x_e - x_d$ for $d, e \in \Omega_k$, so

removing the roots leaves the weights of N' and some zero weights, which do not contribute to r_λ.

For $\lambda \in \mathbb{Z}^\Omega$, let $\mathcal{W}_\lambda$ be its stabilizer in $\mathcal{W}$. Since $\mathcal{B}'$ is $\mathcal{W}$-stable, $w(r_\lambda) = r_{w\lambda}$ for $w \in \mathcal{W}$. For $p \in S^{\mathcal{W}_\lambda}$, the *dressed monopole* $\mathsf{S}_\lambda(p) = \sum_{w\in\mathcal{W}/\mathcal{W}_\lambda} w(p\, r_\lambda)$ lies in A and has the charge of λ. Products are computed with

$$r_\lambda r_\mu = \Big(\prod_{\beta\in\mathcal{B}'} \beta^{(-\langle\beta,\lambda\rangle)_+ + (-\langle\beta,\mu\rangle)_+ - (-\langle\beta,\lambda+\mu\rangle)_+} \Big)\, r_{\lambda+\mu}, \tag{3.1}$$

whose prefactor is a polynomial. Every element of A is a sum of dressed monopoles, one for each $\mathcal{W}$-orbit of coweights.

Degrees. We grade by half the cohomological degree, so each x_d has degree 1. In the grading of [BFN18], halved, the class r_λ has degree $\frac{1}{2}\sum_{\beta\in\mathcal{B}'} |\langle\beta,\lambda\rangle|$. If λ has charge zero, this is

$$\deg r_\lambda = \sum_{\beta\in\mathcal{B}'} (-\langle\beta,\lambda\rangle)_+,$$

the degree of the prefactor of r_λ.

3.3. **The multipartition filtration.** Fix a multipartition m. Split each Ω_k into blocks, one for each part D at k, of size s_D, and choose a *first element* d_D in each block. Let $s \in \mathbb{Z}^\Omega_{\geq 0}$ be s_D at d_D and 0 elsewhere, and put $\lambda_m = s - \mathbf{1}$, where $\mathbf{1} \in \mathbb{Z}^\Omega$ has all entries 1. So λ_m is $s_D - 1$ at the first element of the block D and -1 at its other elements. Listing the elements of a block with its first element first, the entries of λ_m on D are

$$\lambda_m|_D = (\underbrace{s_D - 1, -1, -1, \ldots, -1}_{s_D \text{ entries}}),$$

and λ_m is the concatenation of these vectors over all blocks D. For example, for $m = (3,1)$ at a vertex with $v_k = 4$ it is $(2,-1,-1,0)$. It has charge zero, and its $\mathcal{W}$-orbit does not depend on the choices. The blocks also realize M_m, with $T \subseteq M_m$. Let $\epsilon\colon S \to S_m$ be restriction to $Z_m \subseteq T$, which sets $x_d = x_D$ for d in the block D. For $m = m_G$ we write λ_G, S_G and J_G.

Let $F_L \subseteq B$ be the span of the classes $[\mathsf{S}_{\lambda_m}(p)]$ with $l(m) \leq L$ and $p \in S^{\mathcal{W}_{\lambda_m}}$. It does not depend on the choice of blocks and first elements: another choice replaces λ_m by some $w\lambda_m$, and $\mathsf{S}_{w\lambda}(wp) = \mathsf{S}_\lambda(p)$.

Proposition 3.6. *(1) $F_{|\mathbf{v}|} = B$ and $F_{k_0-1} = 0$.*
(2) For $p \in S^{\mathcal{W}_{\lambda_m}}$, the class $[\mathsf{S}_{\lambda_m}(p)] \in B$ depends only on $\epsilon(p)$, and it has degree $\deg p + \delta_m$.
(3) If $\epsilon(p) \in J_m$, then $[\mathsf{S}_{\lambda_m}(p)] \in F_{l(m)-1}$.
So $\epsilon(p) \mapsto [\mathsf{S}_{\lambda_m}(p)]$ induces a surjection $\bigoplus_{l(m)=L} (S_m/J_m)^{\mathcal{W}_m}\langle\delta_m\rangle \twoheadrightarrow F_L/F_{L-1}$.

Proof. (1) If every part of m has size 1, then every element of Ω is a first element and $s = \mathbf{1}$, so $\lambda_m = 0$, $\mathcal{W}_{\lambda_m} = \mathcal{W}$ and $\mathsf{S}_0(p) = p$ for $p \in S^{\mathcal{W}} = R$. So $F_{|\mathbf{v}|}$ contains the image of R, which is B, as shown at the start of this section. Every multipartition has at least one part at each vertex of Q_0^+, so $l(m) \geq k_0$ and $F_{k_0-1} = 0$.

(2) *The stabilizer.* Write $H = \mathcal{W}_{\lambda_m}$. It preserves the set of first elements of each Ω_k, on which it acts by the group H_1 of permutations of the first elements of parts of equal size at the same vertex; under $d_D \mapsto D$, H_1 is $\mathcal{W}_m$. On the non-first elements of each Ω_k it acts by the full symmetric group, and H is the product of H_1 and these symmetric groups.

Generators of S^H. Let $S^{\mathrm{f}} = \mathbb{C}[x_{d_D} : D \in P_m]$ be the polynomials in the first elements, and for $k \in Q_0^+$ and $j \geq 1$ let

$$P_{k,j} = \sum_{d\in\Omega_k} x_d^j \in R, \qquad P^m_{k,j} = \sum_{D\colon k(D)=k} s_D\, x_{d_D}^j \in (S^{\mathrm{f}})^{H_1}.$$

By (I8)(b), S^H is generated over $(S^{\mathrm{f}})^{H_1}$ by the $P_{k,j}$.

A relation from derivations. It suffices to show that multiplying the dressing by $P_{k,j}$ or by $P_{k,j}^m$ gives the same class in B. This comes from a derivation of A that acts by zero on B. For each k and $j \geq 1$, let $\eta_{k,j}$ be the map

$$\eta_{k,j} \colon p\, e^\lambda \mapsto p\, e^\lambda \sum_{d \in \Omega_k} \lambda_d x_d^j$$

on $\mathrm{Frac}(S)[\mathbb{Z}^\Omega]$. It is a derivation, since the multiplier is additive in λ. It commutes with $\mathcal{W}$, since $\mathcal{W}$ permutes each Ω_k, and it multiplies each r_λ by a polynomial. So it preserves A and its charge components, and as it is a derivation it preserves the ideal I_B. So it induces a map on $B = A/I_B$, and this map is zero. Indeed, $R \to B$ is onto, so every $a \in A$ is $r + i$ with $r \in R$ and $i \in I_B$. Now $\eta_{k,j}(r) = 0$, since r has $\lambda = 0$, and $\eta_{k,j}(i) \in I_B$. So $[\eta_{k,j}(a)] = 0$ in B for every $a \in A$.

Take $a = \mathsf{S}_{\lambda_m}(p) = \sum_{w \in \mathcal{W}/H} w(p\, r_{\lambda_m})$ with $p \in S^H$. The multiplier of $\eta_{k,j}$ at $w\lambda_m$ is $w(\sigma_{k,j})$, where $\sigma_{k,j}$ is its value at λ_m:

$$\sigma_{k,j} = \sum_{d \in \Omega_k} (\lambda_m)_d x_d^j = -P_{k,j} + P_{k,j}^m.$$

It is H-invariant, so $\eta_{k,j}(a) = \mathsf{S}_{\lambda_m}(\sigma_{k,j}\, p)$. Since $[\eta_{k,j}(a)] = 0$, for all $p \in S^H$ we have

$$[\mathsf{S}_{\lambda_m}(P_{k,j}\, p)] = [\mathsf{S}_{\lambda_m}(P_{k,j}^m\, p)]. \tag{3.2}$$

Dependence on $\epsilon(p)$. Let $\epsilon^{\mathrm{f}} \colon S^{\mathrm{f}} \to S_m$ be the restriction of ϵ, the isomorphism with $x_{d_D} \mapsto x_D$. The polynomials $P_{k,j}$ and $P_{k,j}^m$ have the same image $\sum_D s_D x_D^j$ under ϵ, so applying (3.2) repeatedly shows that $[\mathsf{S}_{\lambda_m}(p)]$ depends only on $\epsilon(p)$. Also $\epsilon(S^H) = S_m^{\mathcal{W}_m}$.

Degree. The element $\mathsf{S}_{\lambda_m}(p)$ is a sum of $\mathcal{W}$-translates of $p\, r_{\lambda_m}$, and λ_m has charge zero, so by §3.2 its degree is $\deg p + \sum_{\beta \in \mathcal{B}'} (-\langle \beta, \lambda_m \rangle)_+$. Counting the weights of N' arrow by arrow shows that this sum is $\dim N' - \dim N'^{\xi_m} = \delta_m$.

(3) For each nonempty set I of parts, we split λ_m into two coweights, one of nonzero charge, and expand their product, which vanishes in B. Its leading term involves q_I, and the other terms lie in $F_{l(m)-1}$.

The cut relation. Fix a nonempty set $I \subseteq P_m$ of parts. Let $b \in \mathbb{Z}^\Omega_{\geq 0}$ agree with s at the first elements of the parts in I and vanish elsewhere, and put $a = s - b$. So a records the parts outside I, b records the parts in I, and $\lambda_m = (a - \mathbf{1}) + b$. The charge of b is nonzero, since its entries are ≥ 0 and $s_D > 0$ for $D \in I$. So for $p_a \in S^{\mathcal{W}_a}$ and $p_b \in S^{\mathcal{W}_b}$, the element $\mathsf{S}_b(p_b)$ lies in I_B, and in B we have

$$[\mathsf{S}_{a-\mathbf{1}}(p_a)\, \mathsf{S}_b(p_b)] = 0. \tag{3.3}$$

(Note $\mathcal{W}_{a-\mathbf{1}} = \mathcal{W}_a$.) We expand this product. Since $w(r_\lambda) = r_{w\lambda}$,

$$\mathsf{S}_{a-\mathbf{1}}(p_a)\, \mathsf{S}_b(p_b) = \sum_{w \in \mathcal{W}/\mathcal{W}_a} \sum_{w' \in \mathcal{W}/\mathcal{W}_b} w(p_a)\, w'(p_b)\, r_{wa-\mathbf{1}}\, r_{w'b},$$

and by (3.1), the term of (w, w') is a polynomial times $r_{t-\mathbf{1}}$, where $t = wa + w'b$. By §3.2, grouping the terms by the $\mathcal{W}$-orbit of $t - \mathbf{1}$ writes the product as a sum of dressed monopoles, one for each orbit.

Which orbits occur. Since $t \geq 0$ and $\sum_{d \in \Omega_k} t_d = v_k$, the nonzero entries of t on each Ω_k, in decreasing order, form a partition of v_k. Let m' be the multipartition they define; then $t - \mathbf{1}$ is conjugate to $\lambda_{m'}$. The supports of a and b are disjoint, with $l(m)$ elements in all, so $l(m')$ is $l(m)$ minus the number of common elements of the supports of wa and $w'b$. The orbits coming from terms with overlapping supports therefore give dressed monopoles in $F_{l(m)-1}$. If the supports are disjoint, then the nonzero entries of t at k are those of a and b together, which are the parts of m at k. So $m' = m$, and all these terms lie on the orbit of λ_m. That orbit contributes $\mathsf{S}_{\lambda_m}(f)$, where f is the coefficient of r_{λ_m}, which comes from the pairs $(wa, w'b)$ with $wa + w'b = s$.

The coefficient of r_{λ_m}. These pairs are the (ha, hb) for $h \in H$, and they are indexed by H/H_I, where $H_I = \mathcal{W}_a \cap \mathcal{W}_b$. Write $r_{a-\mathbf{1}} r_b = g\, r_{\lambda_m}$, with g the prefactor of (3.1). Since p_a is $\mathcal{W}_a$-invariant and p_b is $\mathcal{W}_b$-invariant, the pair (ha, hb) contributes $h(p_a p_b\, g)\, r_{\lambda_m}$, so $f = \sum_{h\in H/H_I} h(p_a p_b\, g)$. Comparing the exponents in g with Definition 3.2, $g = \pm \tilde{q}_I$, where $\tilde{q}_I = (\epsilon^{\mathrm{f}})^{-1}(q_I)$ is q_I written in the first elements. By (3.3), we have shown

$$\Big[\mathsf{S}_{\lambda_m}\Big(\textstyle\sum_{h\in H/H_I} h(\tilde{q}_I\, p_a p_b)\Big)\Big] \in F_{l(m)-1}. \tag{3.4}$$

Choice of dressings. Averaging (3.4) over suitable dressings p_a and p_b gives $[\mathsf{S}_{\lambda_m}(\mathrm{Av}_H(\tilde{q}_I y))] \in F_{l(m)-1}$ for every $y \in S^{\mathrm{f}}$, and the images $\epsilon(\mathrm{Av}_H(\tilde{q}_I y)) = \mathrm{Av}_{\mathcal{W}_m}(q_I\, \epsilon(y))$ span $\mathrm{Av}_{\mathcal{W}_m}(J_m) = J_m^{\mathcal{W}_m}$.

Now let $p \in S^H$ with $\epsilon(p) \in J_m$. By (2), $\epsilon(p) \in S_m^{\mathcal{W}_m}$, so $\epsilon(p) \in J_m^{\mathcal{W}_m}$, and $\epsilon(p) = \epsilon(p')$ for a combination p' of the $\mathrm{Av}_H(\tilde{q}_I y)$. By (2), $[\mathsf{S}_{\lambda_m}(p)] = [\mathsf{S}_{\lambda_m}(p')] \in F_{l(m)-1}$.

Finally, by (2), $\epsilon(S^H) = S_m^{\mathcal{W}_m}$, and $(S_m/J_m)^{\mathcal{W}_m} = S_m^{\mathcal{W}_m}/J_m^{\mathcal{W}_m}$ by (I8)(a). So (2) and (3) give a well-defined map $(S_m/J_m)^{\mathcal{W}_m}\langle\delta_m\rangle \to F_L/F_{L-1}$ for each m with $l(m) = L$, of degree zero. The sum of these maps is onto, since F_L is spanned by F_{L-1} and the $[\mathsf{S}_{\lambda_m}(p)]$ with $l(m) = L$. □

This is the argument of [Ope26, Prop. 5.2].

3.4. **The induced filtration.** Put $F'_L = \Phi(F_L) \subseteq H^*(X)$. As Φ is onto, these subspaces filter $H^*(X)$, with $F'_{k_0-1} = 0$ and $F'_{|\mathbf{v}|} = H^*(X)$, and Φ induces surjections $F_L/F_{L-1} \to F'_L/F'_{L-1}$.

Lemma 3.7. *Suppose that for every L the composite*

$$\bigoplus_{l(m)=L} (S_m/J_m)^{\mathcal{W}_m}\langle\delta_m\rangle \twoheadrightarrow F_L/F_{L-1} \twoheadrightarrow F'_L/F'_{L-1} \tag{3.5}$$

of the surjection of Proposition 3.6 with the map induced by Φ is injective. Then Theorem 3.1 holds.

Proof. Both maps in (3.5) are then isomorphisms, so $\dim B = \sum_L \dim F_L/F_{L-1} = \sum_L \dim F'_L/F'_{L-1} = \dim H^*(X)$, and the surjection Φ is an isomorphism. As Φ preserves degrees, the Poincaré polynomial of X is $\sum_m q^{\delta_m} h_m(q)$. □

The first case is $L = k_0$. The only multipartition with k_0 parts is m_G, with one part at each vertex, for which $\mathcal{W}_{m_G}$ is trivial and ϵ maps $S^{\mathcal{W}_{\lambda_G}}$ onto S_G. As $F_{k_0-1} = 0$, Proposition 3.6(2) and (3) give a well-defined map

$$\psi\colon (S_G/J_G)\langle\delta_{m_G}\rangle \to H^*(X), \qquad \psi(\epsilon(p)) = \Phi[\mathsf{S}_{\lambda_G}(p)] \quad \text{for } p \in S^{\mathcal{W}_{\lambda_G}},$$

and this is the composite (3.5) for $L = k_0$. The other values of L reduce to this case for other quivers, by restriction to Levi subgroups (Proposition 3.10), and in §3.6 we show that ψ is injective (Corollary 3.14).

3.5. **Levi subgroups.** A Levi subgroup $M \supseteq T$ is given by a decomposition of each Ω_k into blocks. The gauge theory of M acting on $\mathfrak{m} \oplus N'$ is the quiver gauge theory of the following quiver Q_M:

- its vertices are the blocks, with dimension the size of the block;
- the block (k, D) has framing W_k and one loop, the (D, D) block of ℓ_k;
- for each arrow $k \to l$ of Q' and all blocks (k, D) and (l, E), there is an arrow $(k, D) \to (l, E)$.

Keeping only the block-diagonal part $\mathfrak{m}$ of the chosen loops, rather than all of $\mathfrak{g}$, gives Q_M a loop at every vertex, so Q_M satisfies the hypotheses of this section; the off-diagonal part is recovered in Lemma 3.9. We write $A(M)$, $B(M)$, $X(M)$, Φ_M and κ_M for its Coulomb branch, B-algebra, quiver variety at determinant stability, comparison map and Kirwan map. Its Weyl group $\mathcal{W}_M \subseteq \mathcal{W}$ is the product of the symmetric groups of the blocks. We write $\mathsf{S}^M_\lambda(p)$ for its dressed monopoles, which are sums over $\mathcal{W}_M/\mathcal{W}_{M,\lambda}$, where $\mathcal{W}_{M,\lambda} = \mathcal{W}_M \cap \mathcal{W}_\lambda$.

For $M = M_m$, built from the blocks used to define λ_m, the multipartition of Q_M with one part per block has coweight λ_m, graph Γ_m, and data S_m and J_m.

Lemma 3.8 (restriction)**.** *(1) Proposition 3.5 for G and for M gives $A \subseteq A(M)$, with the same r_λ. Followed by projection to the part of M-charge zero, this induces a ring homomorphism $\mathrm{res}_M \colon B \to B(M)$ with $\mathrm{res}_M[f] = [f]$ for $f \in R$.*
(2) If $l(m)$ is less than the number of blocks of M, then $\mathrm{res}_M[\mathsf{S}_{\lambda_m}(p)] = 0$.
(3) $\mathrm{res}_{M_m}[\mathsf{S}_{\lambda_m}(p)] = [\mathsf{S}^{M_m}_{\lambda_m}(p)]$, and this depends only on $\epsilon(p)$. If $m' \neq m$ and $l(m') = l(m)$, then $\mathrm{res}_{M_m}[\mathsf{S}_{\lambda_{m'}}(p)] = 0$.

Proof. (1) The quiver Q_M has one chosen loop at each block, and removing these loops leaves the M-module N': the arrows of Q_M between blocks are the block components of the arrows of Q', and its framings are the block components of $\mathrm{Hom}(W, V)$. So Proposition 3.5 for Q_M uses the same torus T, the same weights $\mathcal{B}'$ and the same r_λ, and it gives $A(M) = \big(\bigoplus_\lambda S\, r_\lambda\big)^{\mathcal{W}_M}$. Hence $A \subseteq A(M)$.

The M-charge of r_λ is the vector of the sums of λ over the blocks. For $a \in A$, let a_0 be its component of M-charge zero. Then $\rho(a) = [a_0] \in B(M)$ is a ring homomorphism, and it kills the elements of nonzero G-charge, since the M-charge refines the G-charge. So it kills I_B and induces $\mathrm{res}_M \colon B \to B(M)$. An element $f \in R$ has charge zero, so $\mathrm{res}_M[f] = [f]$.

For (2) and (3), let $\lambda \in \mathbb{Z}^\Omega$ and $p \in S^{\mathcal{W}_\lambda}$. Split $\mathcal{W}/\mathcal{W}_\lambda$ into $\mathcal{W}_M$-orbits. The stabilizer in $\mathcal{W}_M$ of $w\mathcal{W}_\lambda$ is $\mathcal{W}_{M,w\lambda} = \mathcal{W}_M \cap \mathcal{W}_{w\lambda}$, and wp is $\mathcal{W}_{w\lambda}$-invariant, so

$$\mathsf{S}_\lambda(p) = \sum_{w \in \mathcal{W}_M \backslash \mathcal{W} / \mathcal{W}_\lambda} \mathsf{S}^M_{w\lambda}(wp).$$

The term of w has M-charge equal to the block sums of $w\lambda$, so res_M keeps exactly the terms in which all block sums of $w\lambda$ vanish.

Take $\lambda = \lambda_m = s - \mathbf{1}$. Then $w\lambda_m = ws - \mathbf{1}$, and ws is s_D at $w(d_D)$ for each part D and 0 elsewhere. So the sum of $w\lambda_m$ over a block $\beta \subseteq \Omega_k$ is $\sum_{D\colon w(d_D) \in \beta} s_D - |\beta|$. For this to vanish, the parts D with $w(d_D) \in \beta$ must have sizes adding up to $|\beta| \geq 1$, so at least one $w(d_D)$ must lie in β. If $l(m)$ is less than the number of blocks, then for every w some block contains no $w(d_D)$. This proves (2).

For (3), let $M = M_m$, which has $l(m)$ blocks, the block of D having size s_D. Let $l(m') = l(m)$. If $w\lambda_{m'}$ has M-charge zero, then each block contains exactly one of the elements $w(d_{D'})$, for D' a part of m', and that part has the size of the block. So at each vertex the parts of m' are the block sizes, which are the parts of m, and $m' = m$. This proves the last claim. For $m' = m$, the condition says that on each block $w\lambda_m$ has one entry equal to the size of the block minus one and all other entries -1. These coweights form the $\mathcal{W}_M$-orbit of λ_m, so only the double coset of $w = 1$ survives, and $\mathrm{res}_{M_m}[\mathsf{S}_{\lambda_m}(p)] = [\mathsf{S}^{M_m}_{\lambda_m}(p)]$. In Q_{M_m}, λ_m is the coweight of the multipartition with one part per block, whose data are Z_m, S_m and ϵ. So Proposition 3.6(2) for Q_{M_m} shows that this class depends only on $\epsilon(p)$. □

Lemma 3.9 (induction)**.** *Let $X(M)^\circ \subseteq X(M)$ be the open subset where, at each vertex k, the blocks of ℓ_k have pairwise disjoint spectra.*
(1) There is a morphism $\iota_M \colon X(M)^\circ \to X$ with $\iota_M^* \mathcal{V}_k \cong \bigoplus_D \mathcal{V}_{k,D}$, where $\mathcal{V}_{k,D}$ is the tautological bundle of the vertex (k, D) of Q_M. So $\iota_M^* \kappa(f) = \kappa_M(f)|_{X(M)^\circ}$ for $f \in R$.
(2) Restriction $H^*(X(M)) \to H^*(X(M)^\circ)$ is injective.
(3) $\iota_M^* \circ \Phi = \Phi_M \circ \mathrm{res}_M$, as maps from B to $H^*(X(M)) \subseteq H^*(X(M)^\circ)$.

Proof. Write a point of $T^*N = T^*(\mathfrak{g} \oplus N')$ as (ℓ, ℓ^*, n), with $\ell, \ell^* \in \mathfrak{g}$ the components for the chosen loops and $n \in T^*N'$, and a point of $T^*(\mathfrak{m} \oplus N')$ as (ℓ_M, ℓ_M^*, n) with $\ell_M, \ell_M^* \in \mathfrak{m}$. Identify $\mathfrak{g}^*$ with $\mathfrak{g}$ by the trace form, let $\mu' \colon T^*N' \to \mathfrak{g}$ be the moment map of G on T^*N', and let $\mathrm{pr}_\mathfrak{m}$ and $\mathrm{pr}_{\mathfrak{m}^\perp}$ be the block-diagonal and off-block-diagonal parts. Then

$$\mu(\ell, \ell^*, n) = [\ell, \ell^*] + \mu'(n), \qquad \mu_M(\ell_M, \ell_M^*, n) = [\ell_M, \ell_M^*] + \mathrm{pr}_\mathfrak{m} \mu'(n).$$

Suppose the blocks of ℓ_M at each vertex have pairwise disjoint spectra. For blocks $D \neq E$ at k and $\gamma \in \mathfrak{m}^\perp$, the (E,D) block of $[\ell_M, \gamma]$ is $\ell_{k,E}\gamma_{ED} - \gamma_{ED}\ell_{k,D}$. So by (I9), ad_{ℓ_M} is invertible on $\mathfrak{m}^\perp$. Put

$$\tilde{\imath}(\ell_M, \ell_M^*, n) = \big(\ell_M,\ \ell_M^* + \gamma,\ n\big), \qquad \gamma = -\mathrm{ad}_{\ell_M}^{-1}\mathrm{pr}_{\mathfrak{m}^\perp}\mu'(n) \in \mathfrak{m}^\perp.$$

Since $[\ell_M, \ell_M^*] \in \mathfrak{m}$ and $[\ell_M, \gamma] \in \mathfrak{m}^\perp$,

$$\mu\big(\tilde{\imath}(\ell_M, \ell_M^*, n)\big) = [\ell_M, \ell_M^*] + \mathrm{pr}_{\mathfrak{m}}\mu'(n) + \big([\ell_M, \gamma] + \mathrm{pr}_{\mathfrak{m}^\perp}\mu'(n)\big) = \mu_M(\ell_M, \ell_M^*, n).$$

So $\tilde{\imath}$ maps $\mu_M^{-1}(0)$ into $\mu^{-1}(0)$, and it is M-equivariant.

It also preserves stability. By (I5), a point of $\mu^{-1}(0)$ is stable if and only if the subspace $U \subseteq V$ generated by the images of the W_k under the arrows of the doubled quiver of Q is all of V. Likewise for Q_M, with the subspace $U_M = \bigoplus_D U_M^{(D)}$ generated by the images of the framings of the blocks under the arrows of the doubled quiver of Q_M. Let U be the subspace for $\tilde{\imath}(\ell_M, \ell_M^*, n)$. Then $U = U_M$, because the projections onto the blocks are polynomials in ℓ_M, and by (I9) the blocks of γ are combinations of compositions of block components of the arrows of Q', their duals and the framing maps. So $\tilde{\imath}(\ell_M, \ell_M^*, n)$ is stable if and only if (ℓ_M, ℓ_M^*, n) is.

Hence $\tilde{\imath}$ induces a morphism ι_M from $X(M)^\circ$, the quotient by M of the stable points of $\mu_M^{-1}(0)$ with disjoint block spectra, to $X = \mu^{-1}(0)^{\mathrm{st}}/G$. The tautological bundle $\mathcal{V}_k$ on X comes from the G-module V_k, which as an M-module is $\bigoplus_D V^{(D)}$. So $\iota_M^*\mathcal{V}_k \cong \bigoplus_D \mathcal{V}_{k,D}$. The class $\kappa(f)$ is f evaluated at the Chern roots of the $\mathcal{V}_k$, and $f \in R \subseteq S^{\mathcal{W}_M}$, so $\iota_M^*\kappa(f) = \kappa_M(f)|_{X(M)^\circ}$. This gives (1).

For (2), let $\zeta \in \mathfrak{z}(\mathfrak{m})$ be central, with entry ζ_D on the block D. The translation $\tau_\zeta \colon \ell_M \mapsto \ell_M + \zeta$ commutes with M, preserves μ_M, since $[\zeta, \ell_M^*] = 0$, and preserves stability, since it does not change which subspaces are stable under each $\ell_{k,D}$. So τ_ζ acts on $X(M)$, and it is homotopic to the identity through the $\tau_{t\zeta}$, $t \in [0,1]$. Let $\mathfrak{L}_M$ be the core of $X(M)$. By (I10)(b), each block $\ell_{k,D}$ is nilpotent on $\mathfrak{L}_M$. Choose ζ with distinct entries at each vertex. Then on $\tau_\zeta(\mathfrak{L}_M)$ the block $\ell_{k,D}$ has the single eigenvalue ζ_D, so $\tau_\zeta(\mathfrak{L}_M) \subseteq X(M)^\circ$. The composite

$$H^*(X(M)) \to H^*(X(M)^\circ) \to H^*(\tau_\zeta(\mathfrak{L}_M)) \xrightarrow{\tau_\zeta^*} H^*(\mathfrak{L}_M)$$

is the pullback along $\tau_\zeta \colon \mathfrak{L}_M \to X(M)$, which equals restriction to $\mathfrak{L}_M$ because τ_ζ is homotopic to the identity. That restriction is an isomorphism by (I10)(a). So the first map is injective.

For (3), every class of B is $[f]$ for some $f \in R$, since $R \to B$ is onto, and $\Phi[f] = \kappa(f)$. By Lemma 3.8(1), $\mathrm{res}_M[f] = [f]$, and $\Phi_M[f] = \kappa_M(f)$. So by (1), $\iota_M^*\Phi[f] = \kappa_M(f)|_{X(M)^\circ} = \Phi_M(\mathrm{res}_M[f])|_{X(M)^\circ}$, and by (2) restriction to $X(M)^\circ$ is injective. □

For $M = M_m$, the multipartition of Q_{M_m} with one part per block plays the role of m_G: its coweight, data and shift are λ_m, S_m, J_m and δ_m, and the map ψ of §3.4 for Q_{M_m} is

$$\psi_{M_m} \colon (S_m/J_m)\langle\delta_m\rangle \to H^*(X(M_m)), \qquad \psi_{M_m}(\epsilon(p)) = \Phi_{M_m}[\mathsf{S}_{\lambda_m}^{M_m}(p)].$$

Proposition 3.10 (detecting the pieces)**.** *Fix L, and for each multipartition m with $l(m) = L$ take $M_m \supseteq T$ built from the blocks used to define λ_m. The map*

$$\Psi_L \colon F_L \to \bigoplus_{l(m)=L} H^*(X(M_m)), \qquad b \mapsto \big(\Phi_{M_m}(\mathrm{res}_{M_m} b)\big)_m = \big(\iota_{M_m}^*\Phi(b)\big)_m,$$

kills F_{L-1} and factors through F'_L/F'_{L-1}. Its composite with the surjection of Proposition 3.6 is the direct sum over m of the restrictions of the ψ_{M_m} to the $\mathcal{W}_m$-invariants. So if every ψ_{M_m} is injective, then so is (3.5).

Proof. The two expressions for Ψ_L agree by Lemma 3.9(3), and the second shows that Ψ_L factors through Φ. Each M_m has L blocks, so by Lemma 3.8(2), Ψ_L kills F_{L-1}; hence it factors through F'_L/F'_{L-1}. By Lemma 3.8(3), Ψ_L sends $[\mathsf{S}_{\lambda_m}(p)]$ to $\Phi_{M_m}[\mathsf{S}_{\lambda_m}^{M_m}(p)] = \psi_{M_m}(\epsilon(p))$ in

the summand of m, and to 0 in the others. A direct sum of injective maps into different summands is injective. □

3.6. **The socle and the top degree.** By Proposition 3.10, it remains to show that the map ψ of §3.4 is injective, for every quiver with a loop at every vertex. Let $\mathfrak{L} \subseteq X$ be the core, as in (I10), so that $H^*(X) \cong H^*(\mathfrak{L})$. Put $d_{\max} = \dim N' - z - k_0$, the bound $\delta_m + n_m - l(m)$ of Proposition 3.3 for $m = m_G$.

Lemma 3.11 (socle)**.** *The map ψ is R-linear, where R acts on S_G/J_G through ϵ, which is onto, and on $H^*(X)$ through κ. So ψ is injective if it is injective in the top degree of S_G/J_G, which it maps to degree $d_{\max}$.*

Proof. For $f \in R$ and $p \in S^{\mathcal{W}\lambda_G}$ we have $\mathsf{S}_{\lambda_G}(fp) = f\,\mathsf{S}_{\lambda_G}(p)$, as f is $\mathcal{W}$-invariant, so $\psi(\epsilon(f)\,\epsilon(p)) = \kappa(f)\,\psi(\epsilon(p))$. The map $\epsilon\colon R \to S_G$ is onto, since it sends $\sum_{d\in\Omega_k} x_d$ to $v_k x_k$. So $\ker\psi$ is an ideal of S_G/J_G. By (I8)(c), if it is nonzero it meets the socle, which by Proposition 3.3 is the top degree $n_{m_G} - k_0$ of S_G/J_G. The shift is δ_{m_G}, and $\delta_{m_G} + n_{m_G} - k_0 = d_{\max}$ by Proposition 3.3. □

Proposition 3.12. *B vanishes in degrees above $d_{\max}$. In degree $d_{\max}$ it is spanned by the classes $[\mathsf{S}_{\lambda_G}(p)]$, so ψ maps the top degree of S_G/J_G onto $H^{2d_{\max}}(X)$. Also $\dim\mathfrak{L} \le d_{\max}$.*

Proof. By Proposition 3.6, F_L/F_{L-1} is a quotient of $\bigoplus_{l(m)=L}(S_m/J_m)^{\mathcal{W}_m}\langle\delta_m\rangle$. By Proposition 3.3, S_m/J_m vanishes above degree $n_m - l(m)$, so the summand of m vanishes above degree $\delta_m + n_m - l(m) = \dim N' - z - L$. For $L = k_0$ this is $d_{\max}$, and the only multipartition with k_0 parts is m_G, with one part at each vertex. For $L > k_0$ it is less than $d_{\max}$. Since $F_{k_0-1} = 0$ and $F_{|\mathbf{v}|} = B$, B vanishes above degree $d_{\max}$, and its part of degree $d_{\max}$ is that of F_{k_0}, which is spanned by the $[\mathsf{S}_{\lambda_G}(p)]$. As Φ is onto, ψ maps onto $H^{2d_{\max}}(X)$. If $e = \dim\mathfrak{L}$ exceeded $d_{\max}$, then $H^{2e}(X) \cong H^{2e}(\mathfrak{L})$ would be nonzero by (I10)(d). This is degree e in our grading, so the surjectivity of Φ would contradict $B_e = 0$. □

Lemma 3.13. *Let o be an acyclic orientation of Γ_{m_G}, whose vertices are Q_0^+ and ∞, in which ∞ is the only source. Then $\mathfrak{L}$ contains an irreducible constructible subset $\mathfrak{L}_o$, of dimension at least $d_{\max}$, with the following property. For every edge $k \to l$ of o with $k, l \ne \infty$, the arrows of the doubled quiver from V_l to V_k vanish on $\mathfrak{L}_o$, and at every point of $\mathfrak{L}_o$ some arrow from V_k to V_l is nonzero. In particular $\mathfrak{L}_o \not\subseteq \overline{\mathfrak{L}_{o'}}$ for $o \ne o'$.*

Proof. In this proof k and l run over Q_0^+. The parallel edges between two vertices are oriented the same way, since o is acyclic. Choose a total order $\prec$ on Q_0^+ with $l \prec k$ whenever the edges between k and l point from k to l, and bases $e_{k,1}, \dots, e_{k,v_k}$ of the V_k. Order all the basis vectors so that $e_{l,s}$ comes before $e_{k,r}$ if $l \prec k$, and $e_{k,s}$ before $e_{k,r}$ if $s < r$. Let $V_{\le r}$ be the span of the first r basis vectors, and let $B_o = \prod_k B_k \subseteq G$ be the stabilizer of this flag, where B_k is the group of upper triangular matrices in the basis of V_k.

Let $\mathbb{A} \subseteq T^*N$ be the linear subspace where the maps $W \to V$ are arbitrary, the maps $V \to W$ vanish, and every arrow of the doubled quiver maps $V_{\le r}$ into $V_{\le r-1}$ for all r. So:

- the loops of the doubled quiver at k, namely ℓ_k, ℓ_k^* and the c_{kk} loops coming from Q', are strictly upper triangular;
- for $l \prec k$, the c_{kl} arrows of the doubled quiver from V_k to V_l are arbitrary, and the c_{kl} arrows from V_l to V_k vanish.

Hence

$$\dim\mathbb{A} = \sum_k \Big((2 + c_{kk})\tbinom{v_k}{2} + w_k v_k\Big) + \sum_a v_{t(a)} v_{h(a)},$$

the last sum over the arrows a of Q' that are not loops, and $\dim B_o = \sum_k \binom{v_k+1}{2}$. On $\mathbb{A}$, the component of μ at k is strictly upper triangular with zeros just above the diagonal, so μ maps $\mathbb{A}$ into a space of dimension $\sum_k \binom{v_k-1}{2}$.

Let $p_0 \in \mathbb{A}$ be the following point:

- ℓ_k is the regular nilpotent $e_{k,r} \mapsto e_{k,r-1}$, and the other loops are zero;
- if the edges between k and l point from k to l, then one arrow of the doubled quiver from k to l is $e_{k,v_k} \mapsto e_{l,v_l}$; all other components of arrows between different vertices are zero;
- if $w_k > 0$, some vector of W_k maps to e_{k,v_k}, and the other framing maps are zero.

Then $\mu(p_0) = 0$, since $[\ell_k, \ell_k^*] = 0$ and all other terms vanish. Since ∞ is the only source, every k has an incoming edge, either from ∞, which means $w_k > 0$, or from some $l \succ k$. So by descending induction along $\prec$, each e_{k,v_k} lies in the subspace generated by W, and hence so does $V_k = \mathbb{C}[\ell_k]e_{k,v_k}$. So p_0 is stable.

Let Y be an irreducible component of $\mathbb{A} \cap \mu^{-1}(0)$ through p_0. It is cut out of $\mathbb{A}$ by $\sum_k \binom{v_k-1}{2}$ equations, so $\dim Y \geq \dim \mathbb{A} - \sum_k \binom{v_k-1}{2}$. Let $Y^\circ \subseteq Y$ be the subset of stable points at which every ℓ_k is regular nilpotent and, for every pair k, l whose edges point from k to l, some component from V_k to V_l is nonzero. These conditions are open; on $\mathbb{A}$, ℓ_k is nilpotent, and it is regular if and only if it has rank $v_k - 1$. They hold at p_0, so Y° is open and dense in Y. The group B_o preserves $\mathbb{A}$ and $\mu^{-1}(0)$, and it is connected, so it preserves each irreducible component of $\mathbb{A} \cap \mu^{-1}(0)$, and it preserves Y°.

Let $\mathfrak{L}_o \subseteq X$ be the image of Y°, which is irreducible and constructible. Let λ be the cocharacter of G acting on the r-th basis vector by $t^{|\mathbf{v}|+1-r}$. Then $\lambda(t) \cdot p \to 0$ as $t \to 0$ for every $p \in \mathbb{A}$, and $\mathfrak{L}_o \subseteq \mathfrak{L}$ by (I10)(c).

If p and gp both lie in Y°, then $g \in B_o$, since g_k must preserve the kernels of the powers of the regular nilpotent ℓ_k. Stable points have trivial stabilizer, so the fibers of $Y^\circ \to \mathfrak{L}_o$ are free B_o-orbits, and

$$\dim \mathfrak{L}_o = \dim Y^\circ - \dim B_o \geq \dim \mathbb{A} - \textstyle\sum_k \binom{v_k-1}{2} - \dim B_o = d_{\max}.$$

For the last claim, let $o \neq o'$. All edges at ∞ point away from ∞ in both, so there are $k, l \in Q_0^+$ whose edges point from k to l in o and from l to k in o'. The condition that all arrows of the doubled quiver from V_k to V_l vanish is closed and G-invariant. It holds on $\mathfrak{L}_{o'}$, hence on its closure, and it fails at every point of $\mathfrak{L}_o$. □

Corollary 3.14. *The map ψ of §3.4 is injective. If Γ_{m_G} has an acyclic orientation in which ∞ is the only source, then $\dim \mathfrak{L} = d_{\max}$, and the closures of the $\mathfrak{L}_o$ are distinct irreducible components of $\mathfrak{L}$.*

Proof. If Γ_{m_G} has no orientation o as above, then S_G/J_G vanishes in its top degree $n_{m_G} - k_0$, by Proposition 3.3. If it were nonzero, its socle would be nonzero by (I8)(c) and would lie in that degree. So $S_G/J_G = 0$, and there is nothing to prove.

Otherwise, Lemma 3.13 and Proposition 3.12 give $\dim \mathfrak{L} = d_{\max}$. So each $\overline{\mathfrak{L}_o}$ is an irreducible closed subset of $\mathfrak{L}$ of dimension $\dim \mathfrak{L}$, hence a component, and they are distinct by Lemma 3.13. By (I10)(d), $\dim H^{2d_{\max}}(X)$ is at least the number of orientations o, which by Proposition 3.3 is the dimension of the top degree of S_G/J_G. By Proposition 3.12, ψ maps that top degree onto $H^{2d_{\max}}(X)$, so it is injective there, and by Lemma 3.11 it is injective. □

3.7. **Proof of Theorem 3.1.** Fix L. For each multipartition m with $l(m) = L$, Corollary 3.14 for the quiver Q_{M_m} shows that ψ_{M_m} is injective. By Proposition 3.10, the composite (3.5) is injective, and Lemma 3.7 gives the theorem. □

The proof also shows that the surjection of Proposition 3.6 is an isomorphism $F_L/F_{L-1} \cong \bigoplus_{l(m)=L} (S_m/J_m)^{\mathcal{W}_m} \langle \delta_m \rangle$, and that Ψ_L is injective on F_L/F_{L-1}.

Corollary 3.15. *Let $\mathcal{M}_L$ be the set of Levi subgroups $M \supseteq T$ with at least L blocks. Then $F_{L-1} = \bigcap_{M \in \mathcal{M}_L} \ker \operatorname{res}_M$. In particular the* cuspidal part *$\bigcap_{M \neq G} \ker \operatorname{res}_M$ of B is $F_{k_0} \cong (S_G/J_G)\langle \delta_{m_G} \rangle$. This is the shifted cohomology of the hypertoric variety of the center of G acting on $N'^{\xi_{m_G}}$.*

Proof. The inclusion $\subseteq$ is Lemma 3.8(2), since F_{L-1} is spanned by classes $[\mathsf{S}_{\lambda_m}(p)]$ with $l(m) < L$. Conversely, let b lie in the right side, and let L' be minimal with $b \in F_{L'}$. Suppose

$L' \geq L$. Each M_m with $l(m) = L'$ has $L' \geq L$ blocks, so $\Psi_{L'}(b) = 0$, and as $\Psi_{L'}$ is injective on $F_{L'}/F_{L'-1}$, we get $b \in F_{L'-1}$, a contradiction. So $L' \leq L-1$. For the cuspidal part, the Levi subgroups $M \supseteq T$ other than G are those with at least k_0+1 blocks, so the intersection is F_{k_0}. It is isomorphic to $(S_G/J_G)\langle \delta_{m_G}\rangle$ by the proof of Theorem 3.1, as $F_{k_0-1} = 0$, and the last claim is Proposition 3.3. □

Remark 3.16 (Hausel's formula). For quivers with a loop at every vertex, the formula of Theorem 3.1 for all $\mathbf{v}$ at once is equivalent to Hausel's formula [Hau06], given the identity of Remark 3.4(4). [Ope26, §6] uses this in the other direction. We follow [Nak26, §3].

Fix $\mathbf{w}$, and let m run over all tuples $(m^{(k)})_{k\in Q_0}$ of partitions, of any sizes, with $|m| = (|m^{(k)}|)_k$. For partitions put $\langle \mu, \nu\rangle = \sum_{a,b} \min(\mu_a, \nu_b)$, and put

$$d_m = \sum_{a\in Q_1'} \langle m^{(t(a))}, m^{(h(a))}\rangle,$$

which is $\dim N_{Q'}^{\xi_m}$ for $N_{Q'} = \bigoplus_{a\in Q_1'} \operatorname{Hom}(V_{t(a)}, V_{h(a)})$ and $\mathbf{v} = |m|$, by the computation before Definition 3.2. So $\dim N'^{\xi_m} = d_m + \mathbf{w}\cdot l(m)$, where $l(m)$ is now the vector of the numbers of parts at each vertex. Consider the series in variables z_k, $k \in Q_0$,

$$A_{\mathrm{fr}} = \sum_m z^{|m|} \frac{q^{-d_m - \mathbf{w}\cdot l(m)}}{(q)_m}, \qquad A_0 = \sum_m z^{|m|} \frac{q^{-d_m}}{(q)_m}, \qquad H = \sum_m z^{|m|} q^{-d_m-\mathbf{w}\cdot l(m)} h_m(q).$$

- Let $M_\eta = \mu^{-1}(\eta)/\!/G$ be the affine quotient at $\eta = (\mathrm{id}_{V_k})_k$, of dimension $2\delta(\mathbf{v},\mathbf{w})$. For any quiver, Hausel's formula is

$$\sum_{\mathbf{v}} q^{-\delta(\mathbf{v},\mathbf{w})} P_{M_\eta}(q)\, z^{\mathbf{v}} = \frac{\sum_m z^{|m|} q^{-E(m)-\mathbf{w}\cdot l(m)}/(q)_m}{\sum_m z^{|m|} q^{-E(m)}/(q)_m},$$

  in the form of [Ope26, (6.9)], where $E(m) = \sum_{a\in Q_1} \langle m^{(t(a))}, m^{(h(a))}\rangle - \sum_k \langle m^{(k)}, m^{(k)}\rangle$. For our Q, the chosen loops cancel the last sum, so $E(m) = d_m$, and similarly $\delta(\mathbf{v},\mathbf{w}) = \dim N'$. By (I4) with $\theta = \nu_{\det}$, followed by multiplying the cotangent coordinate by i, M_η is homeomorphic to X. So Hausel's formula says $\sum_{\mathbf{v}} q^{-\dim N'} P_X(q)\, z^{\mathbf{v}} = A_{\mathrm{fr}}/A_0$.
- Theorem 3.1 says that this sum is H, since $q^{-\dim N'} q^{\delta_m} = q^{-\dim N'^{\xi_m}}$.
- Multiplying Remark 3.4(4) by $z^{|m|} q^{-d_m - \mathbf{w}\cdot l(m)}$ and summing over m gives $A_{\mathrm{fr}} = A_0 H$, as the term of a pair (m, I) factors into a term of A_0 and a term of H.

Since A_0 has constant term 1, the first two statements are equivalent, given the third.

In Hausel's formula the multipartitions arise in a different way. Hausel's proof uses an arithmetic Fourier transform. In the point count of an unpublished 2004 note of Nakajima, explained in [Nak26, Rem. 3.2], one counts isomorphism classes of framed and of unframed representations of Q over a finite field by Burnside's lemma. The sum over the elements g of $G(\mathbb{F})$ factors over the eigenvalues of g, the framing changes only the factor of the eigenvalue 1, and the term of m in A_{fr} or A_0 is the contribution of the unipotent g of Jordan type m, after substituting $|\mathbb{F}|^{-1}$ for q. By the Krull–Schmidt theorem, the ratio counts the absolutely indecomposable framed representations, and these count the points of M_η [CV04]. In Theorem 6.1 below, m records instead the multiplicities of the eigenvalues of the chosen loops.

4. Deformations and quantization

This section follows [Ope26, §7]. Write $H_F = H_F^*(\mathrm{pt})$ and $H_{F,\hbar} = H^*_{F\times\mathbb{C}_\hbar^\times}(\mathrm{pt})$.

Lemma 4.1 (graded Nakayama). *Let ν be generic, and suppose $\Phi\colon B_0^\nu \to H^*(X_\nu)$ is an isomorphism. Then for every flavor torus F, the maps $\Phi\colon B_F^\nu \to H_F^*(X_\nu)$ and $\Phi\colon B^\nu(A_{F,\hbar}) \to H^*_{F\times\mathbb{C}_\hbar^\times}(X_\nu)$ are isomorphisms.*

Proof. We treat the second map; the first is the same with H_F in place of $H_{F,\hbar}$. Let H^+ be the part of $H_{F,\hbar}$ of positive degree, and let K be the kernel of the surjection Φ of Theorem 2.3. Tensor the exact sequence $0 \to K \to B^\nu(A_{F,\hbar}) \to H^*_{F\times\mathbb{C}^\times_\hbar}(X_\nu) \to 0$ with $\mathbb{C} = H_{F,\hbar}/H^+$. The last term is free over $H_{F,\hbar}$ by (I3), so the sequence stays exact:

$$0 \to K/H^+K \to B^\nu(A_{F,\hbar})/H^+ \to H^*(X_\nu) \to 0.$$

By (I2), the middle term is B^ν_0, and the right map is the isomorphism Φ for F trivial and $\hbar = 0$. So $K = H^+K$. As K is a graded submodule of a quotient of $R_\hbar$, it is bounded below, and graded Nakayama gives $K = 0$; here $R_\hbar \to B^\nu(A_{F,\hbar})$ is onto by Lemma 2.2. □

Lemma 4.2. *The Betti numbers of X_θ do not depend on the generic parameter θ.*

Proof. Let $Z \cong \mathbb{C}^{Q_0}$ be the space of central values of the moment map, and let $\pi\colon \mu^{-1}(Z)/\!/G \to Z$ be the family of affine quotients. As $R\pi_!\mathbb{Q}$ is constructible, there is a Zariski open dense $U \subseteq Z$ over which the compactly supported Betti numbers of the fibers $\mu^{-1}(\zeta)/\!/G$ are constant; U is connected. Let θ be generic. Every θ' in a small real ball around θ is generic, with the same signs of the $\theta' \cdot \mathbf{b}$, so $X_{\theta'} = X_\theta$ by §1.1. By (I4), $X_{\theta'}$ is homeomorphic to $\mu^{-1}(-i\theta')/\!/G$. The points $-i\theta'$ form a real ball in $i\mathbb{R}^{Q_0}$, which is Zariski dense in Z, so some $-i\theta'$ lies in U. Hence the compactly supported Betti numbers of X_θ are those of the fibers over U, for every generic θ. Poincaré duality on the smooth X_θ gives the ordinary Betti numbers. □

The X_θ for generic θ are even diffeomorphic [Nak94, Cor. 4.2], by the same hyperkähler rotation. That paper assumes that the quiver has no loops, but, as [Nak26, Rem. 3.3] points out, the proof of [Nak94, Cor. 4.2] uses only [Nak94, Thm. 4.1(1), (2)] and the rotation of complex structures, and these do not need that assumption.

Abelianization with root denominators. When Q does not have a loop at every vertex, Proposition 3.5 is not available, and we use the abelianization map of [BFN18] directly. Use T, Ω, Ω_k, S and $\mathcal{W}$ as in §3.2, let F be a flavor torus, and put $S_F = H^*_{T\times F}(\mathrm{pt})$. Let $\mathcal{B}$ be the multiset of $T \times F$-weights of N, and define $r_\lambda \in \mathrm{Frac}(S_F)[\mathbb{Z}^\Omega]$ as in §3.2, with $\mathcal{B}$ in place of $\mathcal{B}'$; here $\langle\beta,\lambda\rangle$ pairs the T-part of β with λ. For the rest of this section r_λ and S_λ have the meanings given here. These r_λ multiply by (3.1) with $\mathcal{B}$ in place of $\mathcal{B}'$, so their S_F-span is the abelian Coulomb branch of (T, N) [BFN18, Thm. 4.1]. The abelianization map $(\iota_*)^{-1}$ is an injective ring homomorphism from A_F to $\mathrm{Frac}(S_F)[\mathbb{Z}^\Omega]$ [BFN18, Lemma 5.17], and the charge corresponds to the charge of λ.

Call $\lambda \in \mathbb{Z}^\Omega$ *minuscule* if on each Ω_k its entries take at most two values, differing by one, and put

$$\mathrm{eu}_\lambda = \prod_{k\in Q_0} \prod_{d,e\in\Omega_k,\ \lambda_d>\lambda_e} (x_d - x_e).$$

For minuscule λ and $p \in S_F^{\mathcal{W}_\lambda}$ put

$$\mathsf{S}_\lambda(p) = \sum_{w\in\mathcal{W}/\mathcal{W}_\lambda} w\Big(\frac{p}{\mathrm{eu}_\lambda} r_\lambda\Big).$$

For such λ the orbit Gr^λ is closed, its T-fixed points are the $w\lambda$, and at $\hbar = 0$ the Euler class of its tangent space at $w\lambda$ is $\pm w(\mathrm{eu}_\lambda)$. So by [BFN18, Prop. 6.6], $(\iota_*)^{-1}$ sends the dressed monopole $M_{\lambda,p}$ of (I1) to $\pm\mathsf{S}_\lambda(p)$, with a sign depending only on λ. In the setting of §3.2, the weights of the chosen loops contribute $\pm\mathrm{eu}_\lambda$ to r_λ for minuscule λ, so these $\mathsf{S}_\lambda(p)$ agree up to sign with those defined there.

Proposition 4.3. *For every quiver, $\Phi\colon B_0 \to H^*(X_{\nu_{\det}})$ is an isomorphism.*

Proof. We follow [Ope26, Prop. 7.3]. Write $X = X_{\nu_{\det}}$. The maps $R \to B_0$ and $\kappa\colon R \to H^*(X)$ are onto, by Lemma 2.1 and (I3), and $\Phi[f] = \kappa(f)$. So we must show that their kernels K_B and K_X agree.

Let Q^+ be Q with a loop ℓ_k^+ added at every vertex. Let $\mathbb{C}_u^\times$ scale ℓ_k^+ with weight 1 and its dual ℓ_k^{+*} with weight -1. This commutes with G and preserves μ, since $[t\ell_k^+, t^{-1}\ell_k^{+*}] = [\ell_k^+, \ell_k^{+*}]$, so $\mathbb{C}_u^\times$ is a flavor torus. Write H_u^* for $\mathbb{C}_u^\times$-equivariant cohomology and u for the equivariant parameter, so that $H^*_{G\times\mathbb{C}_u^\times}(\mathrm{pt}) = R[u]$. Let A^+ and B^+ be the Coulomb branch of Q^+ with flavor $\mathbb{C}_u^\times$ at $\hbar = 0$ and its B-algebra, and let $X^+ = X_{\nu_{\det}}(Q^+)$, with Kirwan map $\kappa^+\colon R[u] \to H_u^*(X^+)$. Every vertex of Q^+ carries a loop, so by Theorem 3.1 and Lemmas 4.1 and 2.1, the kernel K^+ of $R[u] \to B^+$ is the kernel of κ^+. Let $\mathfrak{n} \subseteq R[u]$ be the maximal ideal generated by the positive part R_+ of R and by $u-1$. We show that $K_X[u]$, K^+ and $K_B[u]$ have the same localization at $\mathfrak{n}$.

Quiver variety side. Let p be a stable point of the zero fiber of the moment map of Q^+ whose class is fixed by $\mathbb{C}_u^\times$. As G acts freely on stable points, $t\cdot p = \rho(t)\cdot p$ for a unique $\rho(t) \in G$, and ρ is a cocharacter of G. Its conjugacy class is locally constant, so the fixed locus $(X^+)^{\mathbb{C}_u^\times}$ is the disjoint union of the open and closed subsets Y_ρ on which it is the class of ρ. Let n_d, $d \in \Omega$, be the weights of $\mathbb{C}_u^\times$ on the fibers of the tautological bundles over Y_ρ. If $\rho \neq 1$, then $f = \sum_{d\in\Omega} x_d^2$ and $u^2\sum_d n_d^2$ act on $H_u^*(Y_\rho)$ by elements that differ by a nilpotent, and $f - u^2\sum_d n_d^2 \in R[u]$ does not lie in $\mathfrak{n}$, as it is $-\sum_d n_d^2 \neq 0$ modulo $\mathfrak{n}$. So $H_u^*(Y_\rho)_\mathfrak{n} = 0$. If $\rho = 1$, then $t\cdot p = p$, so $\ell^+ = \ell^{+*} = 0$. For such p the moment map and the stability condition of (I5) are those of Q. So $Y_1 = X$, with $\mathbb{C}_u^\times$ acting trivially on X and on its tautological bundles, and κ^+ followed by restriction to Y_1 is κ extended linearly over $\mathbb{C}[u]$.

By the localization theorem [Bri98, Thm. 3], restriction $H_u^*(X^+) \to H_u^*\big((X^+)^{\mathbb{C}_u^\times}\big)$ becomes an isomorphism after inverting u. As u is a unit at $\mathfrak{n}$,

$$H_u^*(X^+)_\mathfrak{n} \cong H_u^*(Y_1)_\mathfrak{n} = (H^*(X)[u])_\mathfrak{n},$$

compatibly with κ^+ and κ. Localization is exact, so $K^+_\mathfrak{n} = K_X[u]_\mathfrak{n}$.

Coulomb side. Use the abelianization above for Q without flavor, extended linearly over $\mathbb{C}[u]$, and for Q^+ with flavor $\mathbb{C}_u^\times$. It embeds $A[u]$ and A^+ in $\mathrm{Frac}(S[u])[\mathbb{Z}^\Omega]$, compatibly with the charge. The weights of Q^+ are those of Q together with the weights $x_e - x_d + u$ of the added loops, for d, e in the same Ω_k, and the root factors eu_λ do not change. So $r_\lambda^+ = g_\lambda r_\lambda$, where

$$g_\lambda = \prod_{k\in Q_0}\prod_{d,e\in\Omega_k} (x_e - x_d + u)^{(\lambda_d - \lambda_e)_+}.$$

The factor g_λ is $\mathcal{W}_\lambda$-invariant, so for minuscule λ the elements $\mathsf{S}_\lambda^+(p)$ of Q^+, defined with r_λ^+ in place of r_λ, are $\mathsf{S}_\lambda^+(p) = \mathsf{S}_\lambda(g_\lambda p)$. By (I1) these generate A^+ over $R[u]$, so $A^+ \subseteq A[u]$. Conversely, $\sigma_\lambda = \prod_{w\in\mathcal{W}/\mathcal{W}_\lambda} w(g_\lambda)$ lies in $R[u]$ and takes the value 1 at $x = 0$, $u = 1$, so $\sigma_\lambda \notin \mathfrak{n}$. The quotient $h_\lambda = \sigma_\lambda/g_\lambda$ is a $\mathcal{W}_\lambda$-invariant polynomial, and as σ_λ is $\mathcal{W}$-invariant,

$$\mathsf{S}_\lambda(p) = \sigma_\lambda^{-1}\,\mathsf{S}_\lambda(g_\lambda h_\lambda p) = \sigma_\lambda^{-1}\,\mathsf{S}_\lambda^+(h_\lambda p).$$

By (I1) the $\mathsf{S}_\lambda(p)$ generate $A[u]$ over $R[u]$, so $A^+_\mathfrak{n} = A[u]_\mathfrak{n}$ inside $\mathrm{Frac}(S[u])[\mathbb{Z}^\Omega]$. The B-algebra is the quotient by the ideal generated by the elements of nonzero charge, which commutes with localization. So $B^+_\mathfrak{n} = (B_0[u])_\mathfrak{n}$, compatibly with the maps from $R[u]_\mathfrak{n}$, whose images are the multiples of e^0 in both. Hence $K^+_\mathfrak{n} = K_B[u]_\mathfrak{n}$.

Conclusion. We have shown $K_B[u]_\mathfrak{n} = K_X[u]_\mathfrak{n}$, and graded Nakayama gives $K_B = K_X$. □

Theorem 4.4. *Let ν be generic. Then $B_F^\nu = B_F$, and the maps $\Phi\colon B_F \to H_F^*(X_\nu)$ and $\Phi\colon B^\nu(A_{F,\hbar}) \to H^*_{F\times\mathbb{C}_\hbar^\times}(X_\nu)$ are isomorphisms. So the maps $R \to B_F^\nu$ and $\kappa\colon R \to H_F^*(X_\nu)$ are surjective with the same kernel, and Theorem 1.1 holds.*

Proof. By Lemma 2.1, $B_F^\nu = B_F$ and $B_0^\nu = B_0$, which does not depend on ν. By Proposition 4.3 and Lemma 4.2, $\dim B_0 = \dim H^*(X_{\nu_{\det}}) = \dim H^*(X_\nu)$. So the surjection $\Phi\colon B_0^\nu \to H^*(X_\nu)$ of §2 is an isomorphism, and Lemma 4.1 gives the rest. For the kernels, the map $R \to B_F^\nu$ is onto by Lemma 2.1, and $\Phi[f] = \kappa(f)$, so its kernel is that of κ. □

5. A second proof of the dimension bound

This section gives a second proof of Proposition 4.3: $\dim B_0 = \dim H^*(X_{\nu_{\det}})$ for every quiver. It adds no loops and does not use §3. Instead, it writes B_0 as a quotient of a space $P_{\mathbf{d}}$: the part of dimension $(\mathbf{v},1)$ of a cohomological Hall algebra of the framed quiver $\hat{Q}$, modulo decomposable elements and the image of one tautological operator. BPS cohomology spans $P_{\mathbf{d}}$, and Botta and Davison identify it with $H^*(X_{\nu_{\det}})$. The argument gives the dimension only, not the Poincaré polynomial of Theorem 3.1.

The proof is conditional. It uses results of Jindal and Neguț [JN26] whose proofs are only sketched there; see Remark 5.3. Throughout, F is trivial, $\hbar = 0$ on the Coulomb side, $X = X_{\nu_{\det}}$, and $B_0 = A_0/I_B$, where I_B is generated by the elements of nonzero charge.

5.1. **The framed quiver.** Let $\hat{Q}$ be Q with one more vertex ∞ and w_i arrows from ∞ to i, as in §1.1, and put $\mathbf{d} = (\mathbf{v},1)$. Let $V_\infty = \mathbb{C}$ and $\hat{G} = G \times \mathrm{GL}(V_\infty)$. After choosing bases of the W_i, the representation space of $\hat{Q}$ of dimension $\mathbf{d}$ is N, with $\mathrm{Hom}(W_i,V_i) = \mathrm{Hom}(V_\infty,V_i)^{w_i}$. Let $\hat{A}$ be the Coulomb branch of $(\hat{G},N)$, at $\hbar = 0$ and without flavor. Its charges lie in $\mathbb{Z}^{Q_0} \times \mathbb{Z}$. The scalars in $\hat{G}$ act trivially on N, and this splits $\hat{A}$, as in [Ope26, (8.4)].

We use the abelianization of §4, as set out before Proposition 4.3, for $\hat{Q}$ and $\mathbf{d}$ as well as for Q and $\mathbf{v}$. Put $\hat{\Omega} = \Omega \sqcup \{\infty\}$, and let $\hat{x}_d$, $d \in \hat{\Omega}$, be the coordinates on the Lie algebra of the diagonal torus $\hat{T}$ of $\hat{G}$. Put

$$x_d = \hat{x}_d - \hat{x}_\infty \quad (d \in \Omega), \qquad x_\infty = \hat{x}_\infty,$$

so that $H^*_{\hat{T}}(\mathrm{pt}) = S[x_\infty]$. Let $\hat{\mathbf{1}} \in \mathbb{Z}^{\hat{\Omega}}$ be the coweight with all entries 1, and put

$$b = \sum_{d\in\Omega} x_d \in R, \qquad D = 1 + \sum_{k\in Q_0} v_k.$$

Write $\hat{r}_\lambda$ and $\hat{\mathsf{S}}_\lambda(p)$ for the elements of §4 defined for $(\hat{T},N)$, and r_λ, $\mathsf{S}_\lambda(p)$ for those of (T,N), written in the variables x_d.

Lemma 5.1. *Identify $e^{(\mu,0)}$ with e^μ for $\mu \in \mathbb{Z}^\Omega$. Then:*

(1) $\hat{r}_{(\mu,k)} = r_{\mu-k\mathbf{1}}\, e^{k\hat{\mathbf{1}}}$ for $(\mu,k) \in \mathbb{Z}^\Omega \times \mathbb{Z}$; in particular $\hat{r}_{\pm\hat{\mathbf{1}}} = e^{\pm\hat{\mathbf{1}}}$;
(2) $\hat{A} = A_0[x_\infty, e^{\pm\hat{\mathbf{1}}}]$, and the map $A_0 \otimes \mathbb{C}[x_\infty, e^{\pm\hat{\mathbf{1}}}] \to \hat{A}$ is an isomorphism;
(3) the part of $\hat{A}$ of charge (γ,k) is $A_0^{(\gamma-k\mathbf{v})}[x_\infty]\, e^{k\hat{\mathbf{1}}}$;
(4) $\sum_{d\in\hat{\Omega}} \hat{x}_d = b + Dx_\infty$.

Proof. The $\hat{T}$-weights of N are the $\hat{x}_e - \hat{x}_d = x_e - x_d$, for the arrows of Q, and the $\hat{x}_d - \hat{x}_\infty = x_d$, for the framing. In the variables x_d these are the T-weights $\mathcal{B}$ of N, and each pairs with (μ,k) as it pairs with $\mu - k\mathbf{1}$. This gives (1), as $e^{(\mu,k)} = e^{(\mu-k\mathbf{1},0)} e^{k\hat{\mathbf{1}}}$. Since V_∞ is a line, the root factors also agree: $\mathrm{eu}_{(\mu,k)} = \mathrm{eu}_{\mu-k\mathbf{1}}$. So $\hat{\mathsf{S}}_{(\mu,k)}(p) = \mathsf{S}_{\mu-k\mathbf{1}}(p)\, e^{k\hat{\mathbf{1}}}$ for $p \in S^{\mathcal{W}_\mu}[x_\infty]$.

A coweight (μ,k) has all entries in $\{0,1\}$ exactly when $k \in \{0,1\}$ and $\mu - k\mathbf{1}$ has all entries in $\{0,1\}$ (if $k=0$) or in $\{-1,0\}$ (if $k=1$); similarly for $\{0,-1\}$. Also $H^*_{\hat{G}}(\mathrm{pt}) = R[x_\infty]$. So (I1), for $\hat{Q}$ and for Q, shows that $\hat{A}$ is generated over $R[x_\infty]$ by the $\mathsf{S}_\nu(p)\, e^{k\hat{\mathbf{1}}}$, with ν minuscule and $k \in \{0,\pm1\}$, and that these generate $A_0[x_\infty, e^{\pm\hat{\mathbf{1}}}]$. This is the first part of (2). The map in (2) is injective because $\mathbb{Z}^{\hat{\Omega}} = \mathbb{Z}^\Omega \times \mathbb{Z}\hat{\mathbf{1}}$ and x_∞ is transcendental over $\mathrm{Frac}(S)$. For (3), the coweight (μ,k) has charge (c,k), where c is the charge of μ, and $\mu - k\mathbf{1}$ has charge $c - k\mathbf{v}$. For (4), $\sum_{d\in\Omega}(x_d + x_\infty) + x_\infty = b + Dx_\infty$. □

Define an algebra homomorphism

$$\Theta\colon \hat{A} \to B_0, \qquad a\, x_\infty^j e^{k\hat{\mathbf{1}}} \mapsto [a]\,(-b/D)^j \qquad (a \in A_0).$$

It is well defined by Lemma 5.1(2). It is an algebra map because the isomorphism there is one of algebras, and Θ is the tensor product of the quotient map $A_0 \to B_0$ with $x_\infty \mapsto -b/D$, $e^{\hat{\mathbf{1}}} \mapsto 1$. Lemma 5.1(3),(4) give:

(Θ1) Θ kills the part of $\hat{A}$ of charge (γ, k) unless $\gamma = k\mathbf{v}$;
(Θ2) $\Theta\big((\sum_{d\in\hat{\Omega}} \hat{x}_d)\, a\big) = 0$ for all $a \in \hat{A}$;
(Θ3) $\Theta\big(p(\hat{x}_d - \hat{x}_\infty : d \in \Omega)\big) = [p]$ for $p \in R$.

The value $-b/D$ is the only one compatible with (Θ2). The corresponding projection $\hat{x} \mapsto \hat{x} - \frac{1}{D}(\sum_d \hat{x}_d)\hat{\mathbf{1}}$ splits off the scalars in the Lie algebra of $\hat{T}$. Since it divides by D, it does not come from a splitting of $\hat{T}$, which is why we use Θ and not a decomposition of $\hat{G}$.

Lemma 5.2. *If $0 \neq \alpha \leq \mathbf{d}$ and $\alpha \neq \mathbf{d}$, write $\alpha = (\gamma, k)$. Then $\gamma \neq k\mathbf{v}$.*

Proof. Here $k \in \{0, 1\}$. If $k = 0$ then $\gamma \neq 0$. If $k = 1$ then $\gamma \leq \mathbf{v}$ and $\gamma \neq \mathbf{v}$. □

5.2. **The loop-nilpotent Hall algebra.** Let $\mathsf{P} = \mathbb{C}[\hbar, u_a : a \in \hat{Q}_1]$, with maximal ideal $\mathfrak{m} = (\hbar, u_a)$. Here $\hbar$ is a new parameter on the Hall algebra side; it becomes the loop rotation parameter of the Coulomb branch, which we then set to 0. The tripled quiver of $\hat{Q}$ has the arrows a of $\hat{Q}$, their reverses a^*, and a loop ω_k at every vertex k. It carries its canonical potential and the action of a torus with weights $-u_a$ on a, $u_a - \hbar$ on a^* and $\hbar$ on ω_k, which preserves the potential [JN26, §3.1]. Let $\mathcal{H}_\mathsf{P} = \bigoplus_{\mathbf{n}} \mathcal{H}_{\mathsf{P},\mathbf{n}}$ be the loop-nilpotent cohomological Hall algebra of [JN26, §3], with coefficients extended to $\mathbb{C}$. It is the equivariant critical cohomological Hall algebra of the tripled quiver, with the ω_k required to be nilpotent. It is graded by dimension vectors $\mathbf{n} \in \mathbb{Z}_{\geq 0}^{\hat{Q}_0}$, with $\mathcal{H}_{\mathsf{P},0} = \mathsf{P}$, and each $\mathcal{H}_{\mathsf{P},\mathbf{n}}$ is a module over $H^*_{G_\mathbf{n}}(\mathrm{pt}) \otimes \mathsf{P} = \mathsf{P}[z_{k,a} : a \leq n_k]^{\mathrm{sym}}$, where $G_\mathbf{n} = \prod_k \mathrm{GL}_{n_k}$. We use the product $F \star G$, the product of [JN26] twisted by the sign $(-1)^{\sum_{a\colon k\to l} n_l m_k}$ for F of dimension $\mathbf{n}$ and G of dimension $\mathbf{m}$, as in [Ope26, (8.1)]. The twist changes a word only by a sign depending on the dimensions of its letters, so it does not change spans of words.

For $g \in \mathsf{P}[z]^{\mathrm{sym}}$ let $e_\mathbf{n}(g) = g \cdot [\omega = 0]_\mathbf{n} \in \mathcal{H}_{\mathsf{P},\mathbf{n}}$ [JN26, (98)]. Under the shuffle map of [JN26, §3] it goes to $g(z)\prod_k \prod_{a,b=1}^{n_k}(z_{k,b} - z_{k,a} + \hbar)$ [JN26, Lemma 3.6]. By definition, $e_\mathbf{n}(hg) = h \cdot e_\mathbf{n}(g)$ for symmetric h. Let $C_\mathbf{n}$ be the action of $\sum_{k,a} z_{k,a} = c_1(\bigotimes_k \det V_k)$, the first Chern class of the total determinant, on $\mathcal{H}_{\mathsf{P},\mathbf{n}}$. It is a derivation:

$$C_{\mathbf{n}+\mathbf{m}}(F \star G) = (C_\mathbf{n} F) \star G + F \star (C_\mathbf{m} G), \tag{5.1}$$

because the total determinant is multiplicative in short exact sequences; [Dav22, Prop. 4.1] proves this vertex by vertex.

Let $\mathcal{H} = \mathcal{H}_\mathsf{P}/\mathfrak{m}\mathcal{H}_\mathsf{P}$, graded by the $\mathcal{H}_\mathbf{n}$, and write $e_\mathbf{n}(g)$ and $C_\mathbf{n}$ also for their images. This specialization is abstract. The shuffle image of $e_\mathbf{n}(g)$ is divisible by $\hbar^{|\mathbf{n}|}$, so it vanishes in the specialized shuffle algebra, but the class of $e_\mathbf{n}(g)$ in $\mathcal{H}$ need not vanish. Put

$$\mathcal{T}_\mathbf{d} = \sum_{\substack{\alpha+\beta=\mathbf{d}\\ \alpha,\beta\neq 0}} \mathcal{H}_\alpha \star \mathcal{H}_\beta, \qquad P_\mathbf{d} = \mathcal{H}_\mathbf{d}/(\mathcal{T}_\mathbf{d} + C_\mathbf{d}\mathcal{H}_\mathbf{d}).$$

By (5.1), $C_\mathbf{d}\mathcal{T}_\mathbf{d} \subseteq \mathcal{T}_\mathbf{d}$.

We use the following inputs.

(H1) *Polynomial generation* [JN26, Thm. 2.4, Lemma 3.6, Prop. 3.7, Thm. 3.5]. $\mathcal{H}_\mathsf{P}$ is generated as a P-algebra by the elements $e_\mathbf{n}(g)$.
(H2) *BPS generation* [JN26, (101)–(110), Cor. 3.15]. There are P-modules $L_\mathbf{n}$, the loop-nilpotent BPS cohomology, and P-linear maps $\beta_\mathbf{n}\colon L_\mathbf{n} \to \mathcal{H}_{\mathsf{P},\mathbf{n}}$, such that $\mathcal{H}_\mathsf{P}$ is generated as a P-algebra by the $C_\mathbf{n}^k\beta_\mathbf{n}(\ell)$, for $k \geq 0$ and $\ell \in L_\mathbf{n}$. This rests on cohomological integrality [DM20, Thm. C], in its equivariant form [DHM23, Thm. 11.7], [Dav22, Thm. 3.4]. The tautological class u of [JN26, (103)] acts as $C_\mathbf{n}$; this is the convention of [Dav22, §3.4, (18)–(19)].

(H3) *Framed BPS cohomology.* If $\mathbf{w} \neq 0$, then $L_{\mathbf{d}}$ is a free P-module of rank $\dim H^*(X)$. Indeed, $L_{\mathbf{n}} \cong \hbar M_{\mathbf{n}}$ as P-modules [JN26, Prop. 3.14], where $M_{\mathbf{n}}$ is the equivariant BPS Lie algebra of the preprojective algebra of $\hat{Q}$ in dimension $\mathbf{n}$ [BD23, Def. 3.6]. Next, $M_{\mathbf{n}} \cong M_{\mathbf{n}}^{\mathrm{noneq}} \otimes \mathsf{P}$ [DHM23, Cor. 11.10]; the group allowed there contains our torus. Finally, $M_{\mathbf{d}}^{\mathrm{noneq}} \cong H^*(X)$, up to a shift of degrees [BD23, Cor. 3.13]. The last result uses framing arrows from ∞ to i, as here, and the stability δ_∞, for which a module of dimension $\mathbf{d}$ is semistable exactly when no proper submodule contains V_∞. So it is semistable exactly when the images of the W_i generate V, which is $\nu_{\det}$-stability by (I5).

Remark 5.3 (status of the inputs). With independent parameters, the genericity conditions of [JN26, §2.2] hold.

(1) [JN26, Thm. 2.4] is a statement about a shuffle algebra, and its proof is only sketched: it is "completely analogous" to the K-theoretic case treated in earlier work of the same authors. The transfer to $\mathcal{H}_{\mathsf{P}}$ uses [JN26, Prop. 3.7] (proof sketched), [JN26, Lemma 3.6] (stated without proof) and the injectivity of the shuffle map [JN26, Thm. 3.5] (proof omitted).

(2) The PBW isomorphism for the loop-nilpotent algebra is [JN26, (109)], stated as an isomorphism of graded vector spaces. That it is induced by the product, which is what generation needs, is implicit. The proof of [JN26, Cor. 3.15] cites the PBW theorem for the full algebra, Thm. 3.10 there, where (109) is needed.

(3) [JN26, Prop. 3.14] has a complete proof, which uses [JN26, Thm. 3.5]. The identification [BD23, Cor. 3.13] is sheaf-theoretic. It combines a comparison of BPS sheaves for different stability conditions [BD23, Prop. 3.4] with the description of the stable framed BPS sheaf as the intersection complex of $X \times \mathbb{A}^1$ [BD23, Prop. 3.11], where $\mathbb{A}^1$ records the common scalar of the loops ω_k. No Kac polynomials, point counts or Hausel's formula enter.

5.3. **The Hall–Coulomb map.** For $\mathbf{n} \leq \mathbf{d}$ let $\lambda_{\mathbf{n}} \in \mathbb{Z}^{\hat{\Omega}}$ be the coweight that is 1 on the first n_k elements of each $\hat{\Omega}_k$ and 0 elsewhere. It is minuscule, with charge $\mathbf{n}$, and $\lambda_{\mathbf{d}} = \hat{\mathbf{1}}$. For g in $\mathbb{C}[z_{k,a} : a \leq n_k]^{\mathrm{sym}}$ write $g(\hat{x}_{\mathbf{n}})$ for g evaluated at the $\hat{x}_d$ with $(\lambda_{\mathbf{n}})_d = 1$. It lies in $S[x_\infty]^{\mathcal{W}_{\lambda_{\mathbf{n}}}}$.

Lemma 5.4 ([Ope26, Lemma 8.1])**.** *Assume (H1). There is an algebra homomorphism* $\Upsilon\colon (\mathcal{H}, \star) \to \hat{A}$ *with the following properties. It sends* $\mathcal{H}_{\mathbf{n}}$ *into the part of charge* $\mathbf{n}$*, and* $e_{\mathbf{n}}(g)$ *to* 0 *if* $\mathbf{n} \not\leq \mathbf{d}$*. For* $\mathbf{n} \leq \mathbf{d}$*,*

$$\Upsilon(e_{\mathbf{n}}(g)) = \epsilon_{\mathbf{n}}\, \hat{\mathsf{S}}_{\lambda_{\mathbf{n}}}\big(g(\hat{x}_{\mathbf{n}})\big),$$

with $\epsilon_{\mathbf{n}} = \pm 1$ *depending only on* $\mathbf{n}$*. In particular* $\Upsilon(e_{\mathbf{d}}(g)) = \epsilon_{\mathbf{d}}\, g(\hat{x})\, e^{\hat{\mathbf{1}}}$.

Sketch. We follow [Ope26, Lemma 8.1], which treats the shuffle algebra; it applies to $\mathcal{H}_{\mathsf{P}}$ through the shuffle map, an algebra homomorphism. Over $\mathrm{Frac}(\mathsf{P})$, Tsymbaliuk's homomorphism from the shuffle algebra to difference operators [Tsy23, Thm. 5.7] gives an algebra homomorphism Υ_{P} from $(\mathcal{H}_{\mathsf{P}}, \star)$ to difference operators in the $\hat{x}_d$, with $D^\lambda f(\hat{x}) = f(\hat{x} + \hbar\lambda) D^\lambda$. It sends $e_{\mathbf{n}}(g)$ to 0 if $\mathbf{n} \not\leq \mathbf{d}$, and otherwise to $\pm E_{\mathbf{n}}(g)$, where

$$E_{\mathbf{n}}(g) = \sum_{w \in \mathcal{W}/\mathcal{W}_{\lambda_{\mathbf{n}}}} w\Big(g(\hat{x}_{\mathbf{n}})\, \frac{\prod_{a\colon k \to l} \prod (\hat{x}_d - \hat{x}_e + u_a)}{\prod_k \prod (\hat{x}_d - \hat{x}_e)}\Big) D^{w\lambda_{\mathbf{n}}} \qquad \text{[Ope26, (8.2)]}.$$

In the numerator $d \in \hat{\Omega}_k$ and $e \in \hat{\Omega}_l$; in the denominator $d, e \in \hat{\Omega}_k$. In both, $(\lambda_{\mathbf{n}})_d = 1$ and $(\lambda_{\mathbf{n}})_e = 0$. The reason is that Tsymbaliuk's map evaluates a shuffle element on strings with step $\hbar$. The factor $\prod(z_{k,b} - z_{k,a} + \hbar)$ vanishes on every string of length at least two, so only minuscule shifts survive. Its diagonal factors $\hbar^{|\mathbf{n}|}$ cancel the normalization of [Ope26, Lemma 8.1].

The coefficients of the $E_{\mathbf{n}}(g)$ lie in $\mathsf{P}[\hat{x}]$, localized at the $\hat{x}_d - \hat{x}_e + r\hbar$ for $d \neq e$ in the same $\hat{\Omega}_k$ and $r \in \mathbb{Z}$. This ring is stable under the shifts, so the difference operators with coefficients in it form a P-algebra $\mathcal{D}_{\mathsf{P}}$. By (H1), $\Upsilon_{\mathsf{P}}(\mathcal{H}_{\mathsf{P}}) \subseteq \mathcal{D}_{\mathsf{P}}$, with no parameter inverted. Reducing

modulo $\mathfrak{m}$ makes the D^λ commute, and turns $\mathcal{D}_{\mathsf{P}} \otimes_{\mathsf{P}} \mathsf{P}/\mathfrak{m}$ into $S[x_\infty][(\hat{x}_d - \hat{x}_e)^{-1}][\mathbb{Z}^{\hat{\Omega}}]$, with $D^\lambda \mapsto e^\lambda$. Call the resulting map Υ.

At $u_a = 0$ the denominator of $E_{\mathbf{n}}(g)$ is $\mathrm{eu}_{\lambda_{\mathbf{n}}}$, and the numerator is, up to a sign depending on $\mathbf{n}$, the product of the weights β of N with $\langle \beta, \lambda_{\mathbf{n}} \rangle = -1$. This is the prefactor of $\hat{r}_{\lambda_{\mathbf{n}}}$. So $\Upsilon(e_{\mathbf{n}}(g))$ is $\pm \hat{\mathsf{S}}_{\lambda_{\mathbf{n}}}(g(\hat{x}_{\mathbf{n}}))$, which is the image of a dressed minuscule monopole [BFN18, Prop. 6.6]. The images of the generators lie in $\hat{A}$, and by (H1) so does the image of $\mathcal{H}$. The charge of $w\lambda_{\mathbf{n}}$ is $\mathbf{n}$. For $\mathbf{n} = \mathbf{d}$ both products are empty and $\mathcal{W}_{\lambda_{\mathbf{d}}} = \mathcal{W}$, so $E_{\mathbf{d}}(g) = g(\hat{x}) D^{\hat{\mathbf{1}}}$. □

5.4. **The bound.**

Theorem 5.5. *Assume (H1), (H2) and (H3). For every quiver Q, loops and multiple arrows allowed, and all $\mathbf{v}$ and $\mathbf{w}$,*

$$\dim B_0 \le \dim H^*(X).$$

Proof. If $\mathbf{v} = 0$, then $B_0 = \mathbb{C}$ and X is a point. If $\mathbf{w} = 0$ and $\mathbf{v} \neq 0$, then X is empty by (I5). Also $\mathbf{1}$ pairs to zero with every weight of N, so $r_{\pm\mathbf{1}} = e^{\pm\mathbf{1}}$, and by (I1) these lie in A_0. They are inverse to each other and have nonzero charge $\pm\mathbf{v}$, so $1 \in I_B$ and $B_0 = 0$. So assume $\mathbf{v} \neq 0$ and $\mathbf{w} \neq 0$, and let

$$\psi = \Theta \circ \Upsilon|_{\mathcal{H}_{\mathbf{d}}} \colon \mathcal{H}_{\mathbf{d}} \to B_0.$$

Step 1: ψ is onto. For $p \in R$, let $g = p(z_{k,a} - z_{\infty,1})$, which is symmetric at each vertex. By Lemma 5.4, $\Upsilon(e_{\mathbf{d}}(g)) = \epsilon_{\mathbf{d}}\, p(\hat{x}_d - \hat{x}_\infty)\, e^{\hat{\mathbf{1}}}$, so $\psi(e_{\mathbf{d}}(g)) = \epsilon_{\mathbf{d}}[p]$ by (Θ3). The map $R \to B_0$ is onto by Lemma 2.1.

Step 2: $\psi(\mathcal{T}_{\mathbf{d}}) = 0$. Let $\alpha + \beta = \mathbf{d}$ with $\alpha, \beta \neq 0$. Then $\Upsilon(\mathcal{H}_\alpha)$ has charge α, which Θ kills by (Θ1) and Lemma 5.2. As Υ and Θ are multiplicative, $\psi(\mathcal{H}_\alpha \star \mathcal{H}_\beta) = 0$.

Step 3: $\psi(C_{\mathbf{d}}\mathcal{H}_{\mathbf{d}}) = 0$. By (H1), $\mathcal{H}$ is spanned by words in the $e_{\mathbf{n}}(g)$. Letters with $\mathbf{n} = 0$ are scalars, and a word of dimension $\mathbf{d}$ with at least two letters of nonzero dimension lies in $\mathcal{T}_{\mathbf{d}}$. So $\mathcal{H}_{\mathbf{d}}$ is spanned by the $e_{\mathbf{d}}(g)$ and $\mathcal{T}_{\mathbf{d}}$. Now $C_{\mathbf{d}} e_{\mathbf{d}}(g) = e_{\mathbf{d}}\big((\sum_{k,a} z_{k,a}) g\big)$, so

$$\psi(C_{\mathbf{d}} e_{\mathbf{d}}(g)) = \epsilon_{\mathbf{d}}\, \Theta\Big(\big(\textstyle\sum_{d \in \hat{\Omega}} \hat{x}_d \big)\, g(\hat{x})\, e^{\hat{\mathbf{1}}} \Big) = 0$$

by (Θ2). Also $C_{\mathbf{d}}\mathcal{T}_{\mathbf{d}} \subseteq \mathcal{T}_{\mathbf{d}}$, which ψ kills by Step 2. So ψ induces a surjection $P_{\mathbf{d}} \to B_0$.

Step 4: the bound. Put $\bar{L}_{\mathbf{d}} = L_{\mathbf{d}}/\mathfrak{m}L_{\mathbf{d}}$. By (H2), and as tensor products are right exact, $\mathcal{H}$ is generated by the images of the $C_{\mathbf{n}}^k \beta_{\mathbf{n}}(\ell)$. The word argument of Step 3 shows that $\mathcal{H}_{\mathbf{d}}$ is spanned by $\mathcal{T}_{\mathbf{d}}$ and the $C_{\mathbf{d}}^k \beta_{\mathbf{d}}(\ell)$, $\ell \in \bar{L}_{\mathbf{d}}$. The terms with $k \ge 1$ lie in $C_{\mathbf{d}}\mathcal{H}_{\mathbf{d}}$, so $\bar{L}_{\mathbf{d}} \to P_{\mathbf{d}}$ is onto. By (H3), $\dim \bar{L}_{\mathbf{d}} = \dim H^*(X)$. Hence

$$\dim B_0 \le \dim P_{\mathbf{d}} \le \dim \bar{L}_{\mathbf{d}} = \dim H^*(X).$$ □

No injectivity is needed anywhere in this proof, and no map is compared with a geometric one.

Corollary 5.6. *Assume (H1), (H2) and (H3). Then Proposition 4.3 holds: for every quiver, $\Phi \colon B_0 \to H^*(X)$ is an isomorphism. For $\mathbf{w} \neq 0$ the surjections $\bar{L}_{\mathbf{d}} \to P_{\mathbf{d}} \to B_0$ of the proof are isomorphisms of vector spaces. With this in place of Proposition 4.3, the proof of Theorem 4.4, and so of Theorem 1.1, goes through without §3.*

Proof. The surjection $\Phi \colon B_0 = B_0^{\nu_{\mathrm{det}}} \to H^*(X)$ of §2 and Theorem 5.5 give $\dim B_0 = \dim H^*(X)$, finite, so Φ is an isomorphism. Then every map in the chain $\bar{L}_{\mathbf{d}} \to P_{\mathbf{d}} \to B_0$ is a surjection between spaces of the same finite dimension. □

Remark 5.7. Compared with the proof through §3:

(1) This proof uses §2, Lemmas 4.1 and 4.2, and inputs (I1)–(I5). It does not use §3, the loops added in Proposition 4.3, or inputs (I6)–(I10).
(2) It needs (H1)–(H3) and [Ope26, Lemma 8.1]. The proof in the rest of the paper does not use Jindal–Neguț at all, because Lemma 2.1 replaces [Ope26, §8].

(3) It gives $\dim H^*(X)$, but not the formula of Theorem 3.1 or the multipartition filtration. It does not equip $\bar{L}_{\mathbf{d}}$ or $P_{\mathbf{d}}$ with the ring structure of B_0, and we have not checked whether the maps $\bar{L}_{\mathbf{d}} \to P_{\mathbf{d}} \to B_0$ respect gradings.

6. The quiver variety side

In the setting of §3, the filtration of Proposition 3.6 and the formula of Theorem 3.1 can be described on X alone. In this section cohomology carries its cohomological grading, so the degrees of §3 are doubled: x_D has degree 2, and $\langle\delta\rangle$ raises degrees by 2δ. Let $\mathfrak{c} = \prod_{k\in Q_0^+} \mathbb{A}^{v_k}$ be the space of tuples of monic polynomials of degrees v_k, and let

$$\chi\colon X \to \mathfrak{c}, \qquad [p] \mapsto \big(\det(t-\ell_k)\big)_k$$

record the characteristic polynomials of the chosen loops. The function $\det(t-\ell_k)$ is G-invariant, so χ is well defined on X. For the scaling action on X, χ is equivariant for an action of $\mathbb{C}^\times$ on $\mathfrak{c}$ with positive weights, contracting $\mathfrak{c}$ to the point $0 = (t^{v_k})_k$. The map χ is not proper, so we prove the purity of $R\chi_*\mathbb{Q}_X$ by hand.

Strata. For a multipartition m, let

$$\pi_m\colon \mathbb{A}^{P_m} \to \mathfrak{c}, \qquad (c_D)_D \mapsto \Big(\prod_{D\colon k(D)=k} (t-c_D)^{s_D}\Big)_k,$$

and let $\mathfrak{c}_m$ be the image of the set where the c_D at each vertex are pairwise distinct. So $\mathfrak{c}_m$ is the locus where the roots at each vertex have multiplicities given by m.

- The map π_m is finite, since the c_D are roots of the image polynomials, and it factors through $\mathbb{A}^{P_m}/\mathcal{W}_m$, where $\mathcal{W}_m$ permutes the c_D for parts of equal size at the same vertex.
- On the set where the c_D at each vertex are distinct, π_m is a Galois cover of $\mathfrak{c}_m$ with group $\mathcal{W}_m$, the cover by labeled roots. So $\mathfrak{c}_m$ is smooth of dimension $l(m)$, and a $\mathcal{W}_m$-module V gives a local system $\mathcal{L}(V)$ on $\mathfrak{c}_m$. For graded V we write $\mathcal{L}(V)$ and $\mathrm{IC}(\overline{\mathfrak{c}_m}, \mathcal{L}(V))$ for the sums over the degrees i of $\mathcal{L}(V^i)[-i]$ and $\mathrm{IC}(\overline{\mathfrak{c}_m}, \mathcal{L}(V^i))[-i]$.
- The closure $\overline{\mathfrak{c}_m} = \pi_m(\mathbb{A}^{P_m})$ is the union of the $\mathfrak{c}_{m'}$ over the coarsenings m' of m, obtained by merging parts at the same vertex, as happens when roots collide. The map $\mathbb{A}^{P_m}/\mathcal{W}_m \to \overline{\mathfrak{c}_m}$ is finite and birational, and its source is normal, so it is the normalization. Only the point $c = 0$ of $\mathbb{A}^{P_m}/\mathcal{W}_m$ lies over $0 \in \mathfrak{c}$.

Let $\mathfrak{c}_{\geq L} = \bigcup_{l(m)\geq L} \mathfrak{c}_m$, the locus where the loops have at least L distinct eigenvalues in all, counting each vertex separately. It is open.

Levi subgroups. For a Levi subgroup $M \supseteq T$, let $\chi_M\colon X(M) \to \mathfrak{c}_M = \prod_\beta \mathbb{A}^{|\beta|}$ record the characteristic polynomials of the blocks $\ell_{k,\beta}$ of the chosen loops, the product over the blocks β. Let $\Delta_M \subseteq X(M)$ be the closed subset where each block of each loop has a single eigenvalue. The group $N_G(M)$ acts on the gauge theory of M, so $N_G(M)/M$ acts on $X(M)$, preserving Δ_M. The *cuspidal part* of $H^*(X(M))$ is the $N_G(M)/M$-module

$$V_M = \ker\big(H^*(X(M)) \to H^*(X(M)\setminus\Delta_M)\big).$$

By Theorem 6.1(3) below, Φ_M identifies it with the cuspidal part of $B(M)$ in the sense of Corollary 3.15. As in Corollary 3.15, $\mathcal{M}_L$ is the set of Levi subgroups $M \supseteq T$ with at least L blocks.

Theorem 6.1. *(1) $R\chi_*\mathbb{Q}_X$ is pure of weight 0, and*

$$R\chi_*\mathbb{Q}_X \cong \bigoplus_m \mathrm{IC}\big(\overline{\mathfrak{c}_m}, \mathcal{L}(V_{M_m})\big)[-l(m)].$$

(2) $H^*(X) \cong \bigoplus_m V_{M_m}^{\mathcal{W}_m}$. For every L, the kernel of restriction $H^*(X) \to H^*(\chi^{-1}\mathfrak{c}_{\geq L})$ is $\bigcap_{M\in\mathcal{M}_L} \ker \iota_M^* = \Phi(F_{L-1})$.

(3) $V_{M_m} = \Phi_{M_m}\big(F_{l(m)}(M_m)\big) \cong (S_m/J_m)\langle\delta_m\rangle$, as graded $\mathcal{W}_m$-modules, where $F_\bullet(M_m)$ is the filtration of Proposition 3.6 for Q_{M_m}.

So the formula of Theorem 3.1 is the decomposition theorem for χ, read at the cone point. The summand of m lives on the closure of the stratum where the eigenvalues of the loops have multiplicities m; this records the centralizers of the semisimple parts of the loops, not their Jordan types. Its multiplicity space is the cohomology of the hypertoric variety of Z_m acting on N'^{ξ_m}, shifted by δ_m, with its $\mathcal{W}_m$-monodromy, and the stalk at 0 keeps the invariants.

We prove Theorem 6.1 after two lemmas. Write $K = R\chi_*\mathbb{Q}_X$ and $K_M = R\chi_{M*}\mathbb{Q}_{X(M)}$, and use the conventions of (I11).

Lemma 6.2 (local model)**.** *Let $\mathfrak{c}_M^\circ \subseteq \mathfrak{c}_M$ be the open set where the block polynomials at each vertex are pairwise coprime, and $\pi_M\colon \mathfrak{c}_M^\circ \to \mathfrak{c}$ the map multiplying them. Then π_M is étale, $X(M)^\circ = \chi_M^{-1}(\mathfrak{c}_M^\circ)$, and $(\iota_M, \chi_M)\colon X(M)^\circ \to X \times_{\mathfrak{c}} \mathfrak{c}_M^\circ$ is an isomorphism. Hence $\pi_M^* K \cong K_M|_{\mathfrak{c}_M^\circ}$.*

Proof. The Jacobian determinant of $(f, g) \mapsto fg$ on pairs of monic polynomials is, up to sign, the resultant of f and g, so π_M is étale. The block polynomials are coprime exactly when the block spectra are disjoint, so $X(M)^\circ = \chi_M^{-1}(\mathfrak{c}_M^\circ)$.

The inverse of (ι_M, χ_M) conjugates a point into block form along the decomposition of each V_k by the factors of $\det(t - \ell_k)$; the off-block-diagonal part of ℓ^* is then the correction term of Lemma 3.9. The last claim is base change along the étale map π_M. □

Lemma 6.3 (stalks)**.** *Let $M = M_m$ and $c \in \mathfrak{c}_m$. Choose labeled roots (c_D) of c, and let $\tilde{c} \in \mathfrak{c}_M^\circ$ have block polynomials $(t - c_D)^{s_D}$, so that $\pi_M(\tilde{c}) = c$.*

(1) There is an isomorphism $K_c \cong H^*(X(M))$ of mixed Hodge structures, and K_c is pure: $H^i(K_c)$ has weight i.
(2) For $b \in H^*(X)$, the image of b in the stalk K_c vanishes if and only if $\iota_M^* b = 0$.

Proof. Let $\mathfrak{s}_M \subseteq \mathfrak{c}_M$ be the slice where the roots of each block polynomial sum to zero (not to be confused with the open set $\mathfrak{c}_M^\circ$), $X(M)_{\mathfrak{s}} = \chi_M^{-1}(\mathfrak{s}_M)$, and $K_M^{\mathfrak{s}} = R\chi_{M*}\mathbb{Q}_{X(M)_{\mathfrak{s}}}$. Translating each block of ℓ_M by a scalar gives isomorphisms $X(M) \cong \mathfrak{z}(\mathfrak{m}) \times X(M)_{\mathfrak{s}}$ and $\mathfrak{c}_M \cong \mathfrak{z}(\mathfrak{m}) \times \mathfrak{s}_M$ compatible with χ_M, so $K_M \cong \mathbb{Q} \boxtimes K_M^{\mathfrak{s}}$. In these coordinates $\tilde{c} = ((c_D), 0)$. The complex $K_M^{\mathfrak{s}}$ is constructible for a stratification by $\mathbb{C}^\times$-stable strata, as χ_M is $\mathbb{C}^\times$-equivariant, so by (I11)(d) restriction $H^*(X(M)_{\mathfrak{s}}) = R\Gamma(\mathfrak{s}_M, K_M^{\mathfrak{s}}) \to (K_M^{\mathfrak{s}})_0$ is an isomorphism. Since π_M is étale, Lemma 6.2 gives

$$K_c \cong (K_M)_{\tilde{c}} \cong (K_M^{\mathfrak{s}})_0 \cong H^*(X(M)_{\mathfrak{s}}) \cong H^*(X(M)).$$

These are isomorphisms of mixed Hodge structures, and $H^*(X(M))$ is pure by (I11)(a), since $X(M)$ is smooth and retracts onto its proper core (I10)(a). This proves (1).

For (2), let $U \ni c$ be a small neighborhood over which π_M has a section through $\tilde{c}$, with image $\tilde{U}$. The image of b in K_c is its restriction to $\chi^{-1}(U)$, which by Lemma 6.2 is $\chi_M^{-1}(\tilde{U}) \subseteq X(M)^\circ$, embedded by ι_M. So it is the restriction of $\iota_M^* b$ to $\chi_M^{-1}(\tilde{U})$. By Lemma 3.9(3) and the surjectivity of Φ, $\iota_M^* b$ lies in $H^*(X(M))$, and restriction from $H^*(X(M))$ to $\chi_M^{-1}(\tilde{U})$ is the isomorphism of (1). □

Proof of Theorem 6.1. Purity. The complex $\mathbb{Q}_X$ is pure of weight 0, so K has weights ≥ 0 by (I11)(a). By Lemma 6.3(1), every stalk $H^i(K_c)$ has weight i, so K has weights ≤ 0. So K is pure, and by (I11)(b) it is a direct sum of shifted simple perverse sheaves.

Supports. Near a point $c \in \mathfrak{c}_m$, Lemma 6.2 and the proof of Lemma 6.3 identify K, étale locally, with $\mathbb{Q} \boxtimes K_M^{\mathfrak{s}}$ near $((c_D), 0)$, for $M = M_m$. The stratum $\mathfrak{c}_m$ corresponds to $\mathfrak{z}(\mathfrak{m}) \times \{0\}$, since near $\tilde{c}$ the block polynomials stay coprime and each has a single root exactly over $\mathfrak{c}_m$. So the cohomology sheaves of K are locally constant on $\mathfrak{c}_m$. They become constant on the cover by labeled roots, so the monodromy factors through $\mathcal{W}_m$. Hence each simple summand of K is $\mathrm{IC}(\overline{\mathfrak{c}_m}, \mathcal{L})$ for some m and some local system $\mathcal{L}$ on $\mathfrak{c}_m$ with monodromy through $\mathcal{W}_m$. By the same argument, $K_M^{\mathfrak{s}}$ is pure and is a direct sum of shifted IC sheaves with $\mathbb{C}^\times$-stable supports. In the chart, a summand of K supported on $\overline{\mathfrak{c}_m}$ becomes $\mathbb{Q}$ boxed with a summand of $K_M^{\mathfrak{s}}$

supported at 0, and the other summands of K through c become $\mathbb{Q}$ boxed with summands of $K_M^{\mathfrak{s}}$ whose support has positive dimension.

Multiplicities. Write $K_M^{\mathfrak{s}} = K_{\mathrm{cusp}} \oplus K'$, with K_{cusp} supported at 0 and K' a sum of shifted IC sheaves whose supports are $\mathbb{C}^\times$-stable of positive dimension, hence contain 0. Restriction from $\mathfrak{s}_M$ to $\mathfrak{s}_M \setminus 0$ kills the global sections of K_{cusp}. It is injective on those of K', by (I11)(c) and (d). So the global sections of K_{cusp} are the kernel of

$$H^*(X(M)_{\mathfrak{s}}) \to H^*\big(X(M)_{\mathfrak{s}} \setminus \chi_M^{-1}(0)\big),$$

which under $X(M) \cong \mathfrak{z}(\mathfrak{m}) \times X(M)_{\mathfrak{s}}$ is the kernel V_M of restriction to $X(M) \setminus \Delta_M$. The deck transformations of the cover by labeled roots are induced by $N_G(M)/M = \mathcal{W}_m$, which permutes the blocks, so the summand of K supported on $\overline{\mathfrak{c}_m}$ is $\mathrm{IC}(\overline{\mathfrak{c}_m}, \mathcal{L}(V_M))[-l(m)]$. The shift $[-l(m)]$ comes from the normalization of perverse sheaves: on $\mathfrak{c}_m$ this summand is $\mathcal{L}(V_M)$ in the degrees of V_M. This proves (1).

Global sections. The summands of K are constructible for the $\mathbb{C}^\times$-stable strata $\mathfrak{c}_m$, so by (I11)(d) their global sections are their stalks at 0. The normalization $\mathbb{A}^{P_m}/\mathcal{W}_m \to \overline{\mathfrak{c}_m}$ is finite and an isomorphism over $\mathfrak{c}_m$, so $\mathrm{IC}(\overline{\mathfrak{c}_m}, \mathcal{L})$ is the pushforward of $\mathrm{IC}(\mathbb{A}^{P_m}/\mathcal{W}_m, \mathcal{L})$, and only 0 lies over 0. So by (I11)(e), the stalk of $\mathrm{IC}(\overline{\mathfrak{c}_m}, \mathcal{L}(V))[-l(m)]$ at 0 is $V^{\mathcal{W}_m}$, and it maps injectively to the stalk at every point of $\mathfrak{c}_m$. So $H^*(X) = R\Gamma(\mathfrak{c}, K) \cong \bigoplus_m V_{M_m}^{\mathcal{W}_m}$, which is the first claim of (2).

Write $b \in H^*(X)$ as $\sum_m b_m$, with b_m in the global sections of the summand of m. A summand of m'' has nonzero stalk at $c \in \mathfrak{c}_m$ only if m is a coarsening of m''. So, by descending induction on $l(m)$, the kernel of restriction to $\chi^{-1}\mathfrak{c}_{\geq L}$ is the sum of the summands with $l(m) < L$, and it is also the set of b whose stalks vanish at every point of $\mathfrak{c}_{\geq L}$. By Lemma 6.3(2), the stalks of b over $\mathfrak{c}_{m'}$ vanish if and only if $\iota_{M_{m'}}^* b = 0$. Every $M \in \mathcal{M}_L$ is conjugate to some $M_{m'}$ with $l(m') \geq L$, so this kernel is $\bigcap_{M \in \mathcal{M}_L} \ker \iota_M^*$. Finally, by Lemma 3.9(3), $\iota_M^* \circ \Phi = \Phi_M \circ \mathrm{res}_M$, and Φ_M is injective by Theorem 3.1 for Q_M. So $\bigcap_{M \in \mathcal{M}_L} \ker \iota_M^* = \Phi\big(\bigcap_{M \in \mathcal{M}_L} \ker \mathrm{res}_M\big) = \Phi(F_{L-1})$ by Corollary 3.15. This proves (2).

Cuspidal parts. Apply (2) to $X(M)$ for $M = M_m$, with $l(m)+1$ in place of L. The vertices of Q_M are the $l(m)$ blocks, so the loops of Q_M have at least $l(m)+1$ distinct eigenvalues in all exactly off Δ_M. So $V_M = \Phi_M(F_{l(m)}(M))$. By Corollary 3.15 for Q_M, whose gauge group is M, with $l(m)$ blocks, $F_{l(m)}(M)$ is the cuspidal part $(S_m/J_m)\langle\delta_m\rangle$ of $B(M)$. The group $N_G(M)/M = \mathcal{W}_m$ permutes the blocks of equal size at each vertex, so it acts on the gauge theory of Q_M, hence on $B(M)$, $H^*(X(M))$ and S_m, and Φ_M and ϵ commute with these actions. So the identification is $\mathcal{W}_m$-equivariant. This proves (3). □

Remark 6.4. For the Jordan quiver with $\mathbf{w} = 1$, χ is the composite $\mathrm{Hilb}^n(\mathbb{C}^2) \to \mathrm{Sym}^n\mathbb{C}^2 \to \mathrm{Sym}^n\mathbb{C}$, every V_{M_m} is one-dimensional, and Theorem 6.1(1) is the push-forward of the decomposition of Göttsche and Soergel for the Hilbert–Chow map.

Remark 6.5. Decompositions over multiplicity strata also appear for the BPS sheaves of preprojective algebras, which for the Jordan quiver are supported on small diagonals [Dav23, Thm. 5.1, Cor. 5.3]. For any quiver, [DHM23, Thm. 1.14] decomposes the cohomology of Nakajima quiver varieties over the BPS Lie algebra; that decomposition is along $X \to \mu^{-1}(0)//G$ rather than along χ. We have not compared these with Theorem 6.1.

7. Status

(1) Theorem 2.3 is announced in [BT26, Thm. E]; we do not give a proof.
(2) We have not checked the following citations against the sources:
- the form of [BFN18, Prop. 3.22] used in (I2), with flavor and loop rotation;
- [HV13] and [GZ83];
- [BS21, Prop. 3.18] in (I4);
- the theorem numbers in the paragraph on known cases (§1.1) and in the remarks after Theorem 6.1, which were taken from the arXiv versions;

- the smoothness statement in (I6)(a), for which we give no theorem number;
- the degree of r_λ in the grading of [BFN18] (§3.2);
- the form with flavor of [BFN18, Thm. 4.1, Lemma 5.17, Prop. 6.6], used before Proposition 4.3; these were checked without flavor against arXiv:1601.03586v8;
- [CG97, §2.6] for the top homology of a compact variety in (I10)(d);
- [KS94, Prop. 3.7.5];
- the numbering in [BBD82];
- that [Nak94, Thm. 4.1(2)], whose proof there refers to Kronheimer, holds for quivers with loops (remark after Lemma 4.2).

(3) The second proof in §5 depends on (H1) and (H2), whose proofs in [JN26] are only sketched (Remark 5.3), and on [Ope26, Lemma 8.1]. The sketch of Lemma 5.4 follows [Ope26, Lemma 8.1]. Its signs and specialization were compared with [Tsy23] only in a machine reading of the preprint, and the formula [Ope26, (8.2)] was not rederived here.

(4) A proof on the quiver variety side that the cuspidal part V_G is $(S_G/J_G)\langle\delta_{m_G}\rangle$ would give the formula of Theorem 3.1 with no input from the Coulomb branch. Hausel's formula and Remark 3.4(4) already give that formula with no such input (Remark 3.16), and Theorem 6.1(2) gives $H^*(X) \cong \bigoplus_m V_{M_m}^{\mathcal{W}_m}$ without using Φ; what is missing is the comparison summand by summand.

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